G50 – 1600
Radial Drill Manual

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1. Summary of the machine

1. 1 GENERAL SAFETY RULES

Operator must read the instruction before operating the machine, and the manager of safety department should assure the operator knows the requirements well.

WARNING! NEVER TOUCH THE CUTTERS AND OTHER DRIVING PARTS WITH HANDS IF THEY ARE STILL RUNNING.

1.1.1 The operation, maintenance and repair of the machine must be done by qualified person who have been trained to have the ability to forecast the potential risks. Only safety-conscious persons who are fully aware of the risks can operate the machine.

1.1.2 To those people who assemble, operate, or maintain the machine must be confirmed to have read and understood the operating instructions.

1.1.3 After stopping the machine, the tool will still run for a period of time due to inertia, do not open the tool guard and touch the tool with your hand before it comes to stop running.

1.1.4 Do not detach or modify any part of the safety guard. The machine should be disconnected from the power supply during maintenance or repair.

1.1.5 Assign special persons to operate, maintain and adjust the machine.

1.1.6 Operate the machine in technically perfect condition. Only the producer or qualified person can completely repair the machine.

1.1.7 Stop the machine immediately if abnormal phenomena appear, check and repair the machine by specialized persons in time.

1.1.8 Disassembly and assembly the machine with the hoisting equipment, which is of enough load capacity.

1.1.9 Observe all safety instructions and warnings attached to the machine, make sure they are always complete and perfectly legible.

1.1.10 Do safety inspection before operating. Ensure the stroke limit switches, stop block and emergency stop button safe and reliable.

1.1.11Reinstall the removed guards and safety devices in place after maintaining.

1.1.12 Do maintenance or adjustment only after switching the machine off and

disconnecting the machine from the power source.

1.1.13 Always keep children away from the machine. Operating personnel age must be at least 18 years.

1.1.14 Do not wear loose clothing, gloves, neckties or jewellery (rings, watches, etc.). Keep the quills and the edges of the work uniform tight. Always be sure to wear safety goggles and wear safety shoes during operation.

1.1.15 Put the long hair into a safety hat during operation no matter the operator is a man or a woman.

1.1.16 It is recommended to wear suitable hearing protection equipment when necessary to reduce the risk of hearing loss.

1.1.17 Provide adequate light around the machine, and keep the perimeter around the machine dry, clean and in good order. In addition, do not place anything near the machine; otherwise it becomes an obstacle during operation.

1.1.18 Do not remove the guards from their working place during operation.

1.1.19 Disconnect the power supply before leaving the machine.

1.1.20 Restart the machine only after the guards and safety devices are replaced to their position and in their good function.

1.1.21 Do not place tools, work pieces, which is not in use on the machine, especially on the moving parts.

1.1.22 Clamp workpiece and tools completely and reliably, take off the wrench from chuck before starting the spindle running.

1.1.23 Stop the machine before adjusting the position of the coolant nozzles.

1.1.24 Do not use compressed air to remove dust and chips etc. deposited on the machine, switchboard, control unit etc.

1.1.25 Both operators and the maintenance men should read the descriptions carefully on the caution plate attached on the machine. They should observe the OPERATION MANUAL in their work. During operation and maintenance, care should be taken so that the caution plate is not dirtied or damaged.

1.1.26 Always bear in mind where the emergency stop button is so that you can operate it without delay in case of an emergency.

1.1.27 Start the machine according to the starting procedures.

1.1.28 Keep your hands away from the moving parts of the machine during operation.

1.1.29 When removing chip iron around the tooling system, avoid taking them out directly by hand, there is cutting hazards. It is safer with proper tool when doing this. Be sure to stop the machine when removing chip from the machine.

1.1.30 Prior to starting test operation, read this manual carefully so as to be familiar with the machine.

1.1.31 Contact the manufacturer if for some reason about some of the OPERATION MANUAL become unreadable.

1.1.32 When install a tool, stop the spindle and the feed in each axis.

1.2 Additional Safety Rules Of Machine

1.2.1 Must read and understand the entire instruction manual before operating the machine.

Warning: Failure to comply with this instruction may cause serious injury.

1.2.2 Always wear approved safety glasses/suit while operating machine tool.

1.2.3 Make sure the machine is properly grounded.

1.2.4 Before operating the machine, remove tie, rings, watches, other jewelry, Keep the quills and the edges of the work uniform tight. Always be sure to wear safety goggles and wear safety shoes during operation. Do not wear gloves while operating machine tool.

1.2.5 Keep the floor around the machine clean and away from scrap materials, oil and grease , etc.

1.2.6 Keep all guards of the machine securely in place at all times when working. For maintenance purpose, use extreme caution and replace the guards immediately after maintenance.

1.2.7 Make sure workpiece and cutter be fitted and clamped well, and make sure the cutter doesn't touch the workpiece before starting the machine.

1.2.8 Must shut off power supply before adjusting or maintaining the machine.

1.2.9 The operator must keep clear-minded when operating the machine and pay attention to what he is doing. It is not allowed to operate the machine when the operator is tired, after drinking or taking medicines.

1.2.10 Use tools properly. Don't force a tool or an attachment to do work which was not designed. Sharp tools should be used. Deformed or dull tools should not be used.

1.2.11 Make certain the motor switch is in OFF position before connecting the machine with power supply.

1.2.12 Never attempt to operate or adjust the machine when the procedure is not understood.

1.2.13 be careful, the handle of the spindle quill rapidly rebound.

1.2.14 It is advised that the operator should change machining position on the table frequently in order to prolong the life of the machine.

1.2.15 Lubricate the machine in time according to the lubricating demand.

1.2.16 Keep the electrical elements clean, do not clean the electrical elements with kerosene or gas.

1.2.17 Be sure that the spindle rotation of the machine comes to rest before changing spindle speed.

1.2.18 It is forbidden to process flammable and explosive metal, for example: pure aluminium, magnesium and so on.

1.2.19 The machine should never be used in flammable, explosive or humid environment

1.2.20 The machine should be disassembled and assembled supported by the lifting mechanism with sufficient bearing capacity.

1.2.21 Always stop the machine before adjusting the position of the cooling nozzle.

1.2.22 Make sure that the operating site should kept well ventilated. It is recommended that ventilation equipment should be provided on the operating site.

1.2.23 Must use the special swinging rings, bolts that manufacturer supplied to swing the machine according to the swing figure.

1.2.24 Must make sure the each clamp part of the machine be in the clamp state and adjust the machine well according to the demand of swing figure, then may swing the machine.

1.2.25 Must check whether every clamp part is reliable, every travel is reliable, grounding wire is reliable, phase sequence of power supply is right, emergency stop

pushbutton is reliable.

1.2.26 Must clamp workpiece and cutter securely, but excessively cut workpiece.

1.2.27 Must clamp the arm, column, headstock securely, otherwise will result in injury.

1.2.28 Select revolving speed and feed on the table of plate according to the machining material and different diameter of drill, otherwise may cause injury to the machine.

1.2.29 The spindle feed handle will turn when the spindle is automatically feeding, and keep away from the handle, otherwise may cause injury to operator.

1.2.30 Keep others away from the work area during the run of machine.

1.2.31 Carefully observe things around the machine before turning the arm, otherwise will result in serious injury.

1.2.32 There are push points between the headstock and stop block, the arm and column, so operators will look to their own safety.

1.2.33 Shut off main supply before repairing the machine.

1.2.34 Shut off power supply of machine before eliminating the malfunction,

1.2.35 Must push the E-stop button on headstock panel in Figure 3.1 when the machine has a fault, then shut off main switch on the door in Figure 4.4. Do not switch on main switch before eliminating the malfunction.

1.2.36 The operation, maintenance and reparation of the machine must be done by qualified person who have been trained to have the ability to forecast the potential risks. Put long hair into a safety hat during operation.

1.2.37 Operator must be familiar with the position of E-stop switch and working method.

1.2.38 Electrical maintenance personnel work as working rules and wear insulated shoes. Do not switch on main switch during the reparation, and hang electrical warning plate on the power switch.

1.2.39 Appointed person is in charge of the key to electrical door and power switch.

1.2.40 Switch off the main switch before opening the electrical door.

1.2.41 User shall add guard according to the size and shape of the workpiece to avoid causing danger that coolant and scrap iron splash.

1.2.42 If dismantle spindle balancing device during the reparation, please be care with spring.

1.2.43 Must not loosen the lifting rope when the machine is not installed and the foundation bolts are tight.

1.2.44 Must not change cutter, check parts, eliminating fault, clean the scrap iron during work.

1.2.45 Must not bump the stoppers during power feed. Must switch off the switch of the power feed when the headstock approaches the stopper.

1.2.46 Must not wear loose clothing, gloves, neckties or jewellery (rings, watches, etc.). Keep the quills and the edges of the work uniform tight.

1.2.47 Kerosene, gas and other flammable liquid are prohibited to clean the machine. compressed air device is prohibited to clean the machine.

1.2.48 Unauthorized persons must not start, operate, maintain the machine and open the electrical door, contact electrical components.

1.2.49 Must not exceed ±180° when turning the arm in horizontal direction.

1.3. Working environment and note of caution symbol

1.3.1 The machine is designed for operating on the site:

—The height above sea level doesn't exceed 1000m.

—The ambient temperature range doesn't exceed 5℃～40℃.

—The relative humidity doesn't exceed 50% within＋40℃ and 90% within ＋20℃.

—Transportation temperature range is within -25℃～+55℃

__ The light in the working environment should not be lower than 500 lux.

1.3.2 Do not use the machine in an environment of electric dirt,

explosion, metal fretted, gas and steam which can destroy insulation .

1.3.3 Do not use the machine in an environment of impact and vibration.

1.3.4 The noise of the machine ≤ 85 dB (A).

1.3.5 Note of caution symbol



:hazardous electrical voltage.



:Imminent danger that will cause serious injury to personnel or damage to machine.

1.4 Structure of the machine

The machine consists of base, column, table, main drive mechanism, coolant, lubrication, electrical equipment and so on.

	Cast iron	Steel
Drilling	ϕ 50	ϕ 50
Tapping	M20×2.5	M16×2



Machine outside drawing

2. Installation and trial run of machine

CAUTION：

Headstock, arm, column were clamped before leaving the factory. Must open the picking box carefully, otherwise it will scratch the paint on the machine surface. After opening the picking box, check all the accessories according to the packing list, check whether there is something wrong or damaged, please inform dealer or manufacturer in time in order to solve it well. Must not connect the power supply to the machine during the swinging and installation, and must not loosen the clamp handles, otherwise it will result in death or serious injury.

2.1 Preparation of ground

In order to make machine run steadily and keep high machining accuracy, the machine must be installed on the concrete foundation. Floor space of the machine is 2700×1300mm, and the depth of groundsill is no less than 500mm. Must consider contiguous equipment, such as safety routeway and wall around the machine before installing the machine. Min. distance in horizontal direction must be at least 800mm

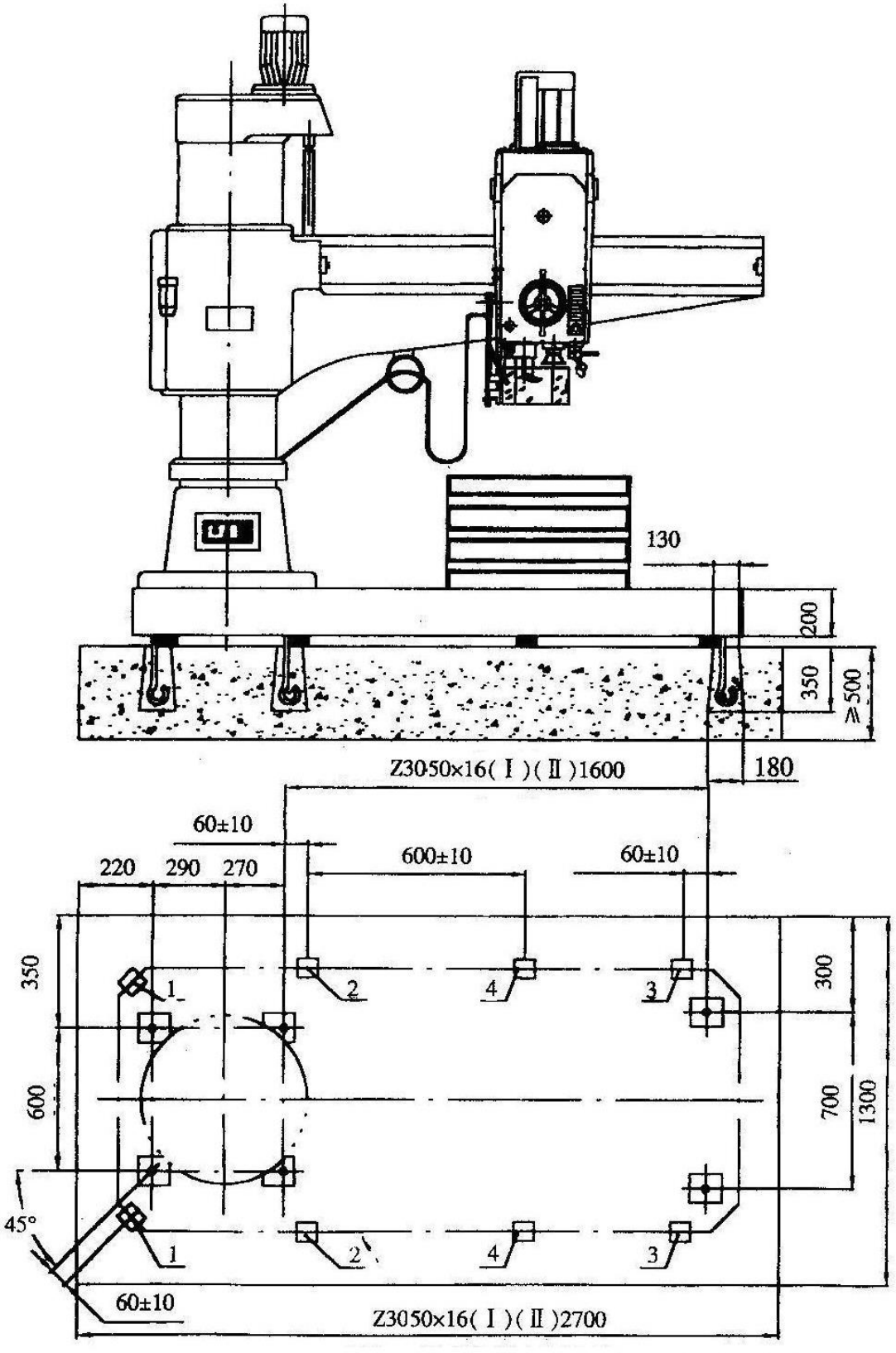

130
200
350
≥500
180
Z3050×16(Ⅰ)(Ⅱ)1600
60±10
600±10
60±10
220
290
270
350
600
300
700
1300
45°
1
2
4
3
60±10
Z3050×16(Ⅰ)(Ⅱ)2700

Figure 2.1

2.2 Machine lifting (Figure 2.2)

Loosen the nut which tighten the working table and machine bed first when opening the box, swivel the working table 90 degrees in level, then move it to the barycenter of the machine along the longitudinal axes, then tighten the working table on the bed with the nut again, attire in the rigging, place some flexible things on the touched surface between rigging and machine to avoid abrading the machine surface, then lift the machine.

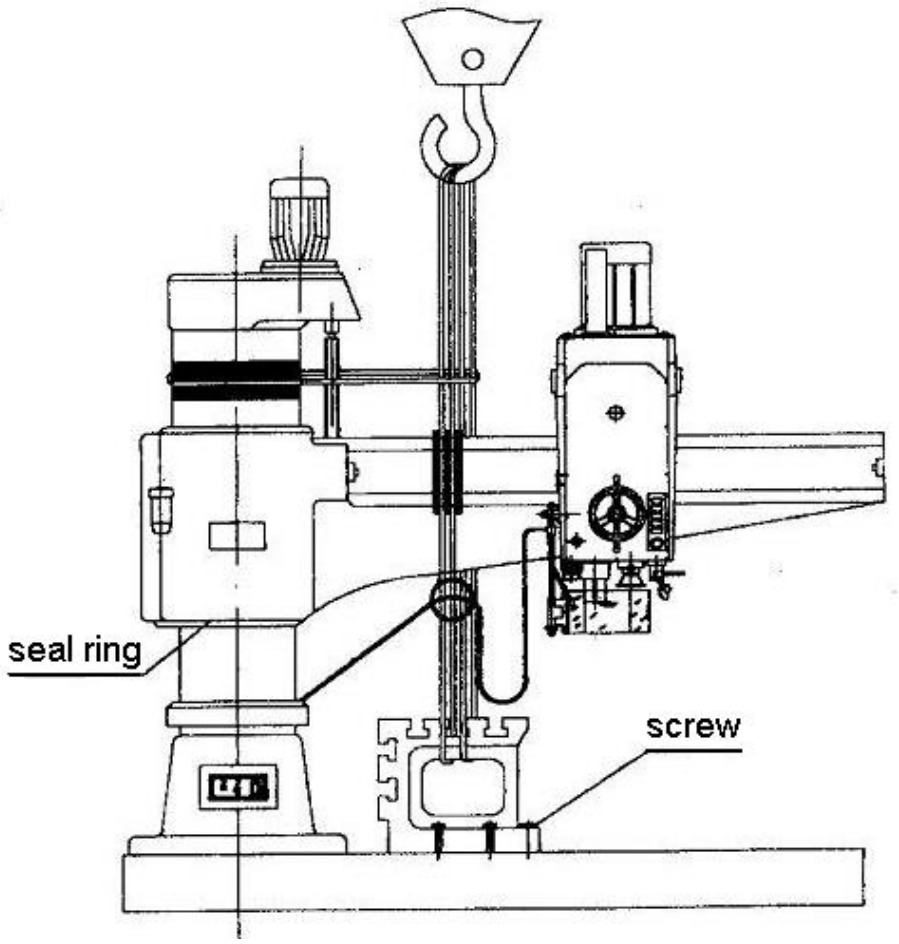


Figure 2.2

2.3 Installation of machine

2.3.1 Put the anchor bolts on the machine base when installing the machine, then put the machine on the groundwork, put the sizing block underlay the machine.

2.3.2 Clean the antirust oil on the base, column, arm, spindle and other surface.

2.3.3 Adjust the level initially: adjust the sequence of the sizing block according to 1,2,3,4 in groundwork picture 2.1.

2.3.4 Remove the locking screw (1) according to picture 2.3, joint the clamp oil pipe.

2.3.5 Remove the locking plate according to picture 2.4.

2.3.6 Electric installation:

2.3.6.1 Lifting motor installation (see picture 2.5)

Fixing the motor in correct way and fix the bolt. Open up the junction box, connect the leads according to the numbers, then cover it and screw down.

2.3.6.2 Connect the leads of main power.

2.3.7 Check the correctness of power supply phase sequence (picture 3.1)：push the button 1-18, check whether the headstock is loose (or push button 1-19, check the headstock and see whether it is clamped); push elevating button 1-3 on the arm，move the arm up by 10mm，check whether it is lifting or not (or push the down button 1-2，check whether it is lowering).

2.3.8 Push the falling button 1-2(picture 3.1)，make the arm come down 10mm，clean the antirust oil on the top of column, move the arm up 10mm again, clean the antirust at the bottom of the column, repeat the work until the oil is rubbed up. Push the button to loosen the headstock, turn the moving handwheel 1-10, move the headstock to the left side, clean the antirust oil on the guideway, then reset and clamp the headstock.

2.3.9 Trickle the lubricating oil on the surface of the column, then move the arm 50mm down, clean and lubricate the bare section again, move the arm 100mm up, clean and lubricate the bare section adequately, so you can lift and lower the arm in a large extend.

Attention: Do not scratch the surface of the column, oiling them according to the lubricating list in picture 2.6 and check the lubricating situation, and immit the

eligible coolant.

2.3.10 move the arm up to the top of the column， then move down to the bottom， check whether the travel button is good.

2.3.11 Adjust level carefully, Adjust the sizing block according to picture 2.1(1,2,3,4) check the level of the cross and longitudinal of the base and make sure the level of the cross and longitudinal of the base is under 0.04/1000.

2.3.12 Adjust the precision of each item, make sure the measurement values do not exceed the values on the accuracy test list. Fill the foundation bolt with cement and sand slurry after the adjustment, fix the sizing block and machine base with cement after they are vulcanized completely. Operator should check with the gradient at last in order to avoid the change of the level.

Remark: 1. Sealing ring under the arm sleeve is put abnormally and openly assemble the end cover on the left of arm are not the correct adjustment position, so when operator install the machine, please do as the correct position.(picture 2.2)

2． Check whether the joint of each oil pipe, oil window, oil mark and other correlative part seep oil, please reassemble the machine if necessary.

3． The oil should not exceed the red mark.

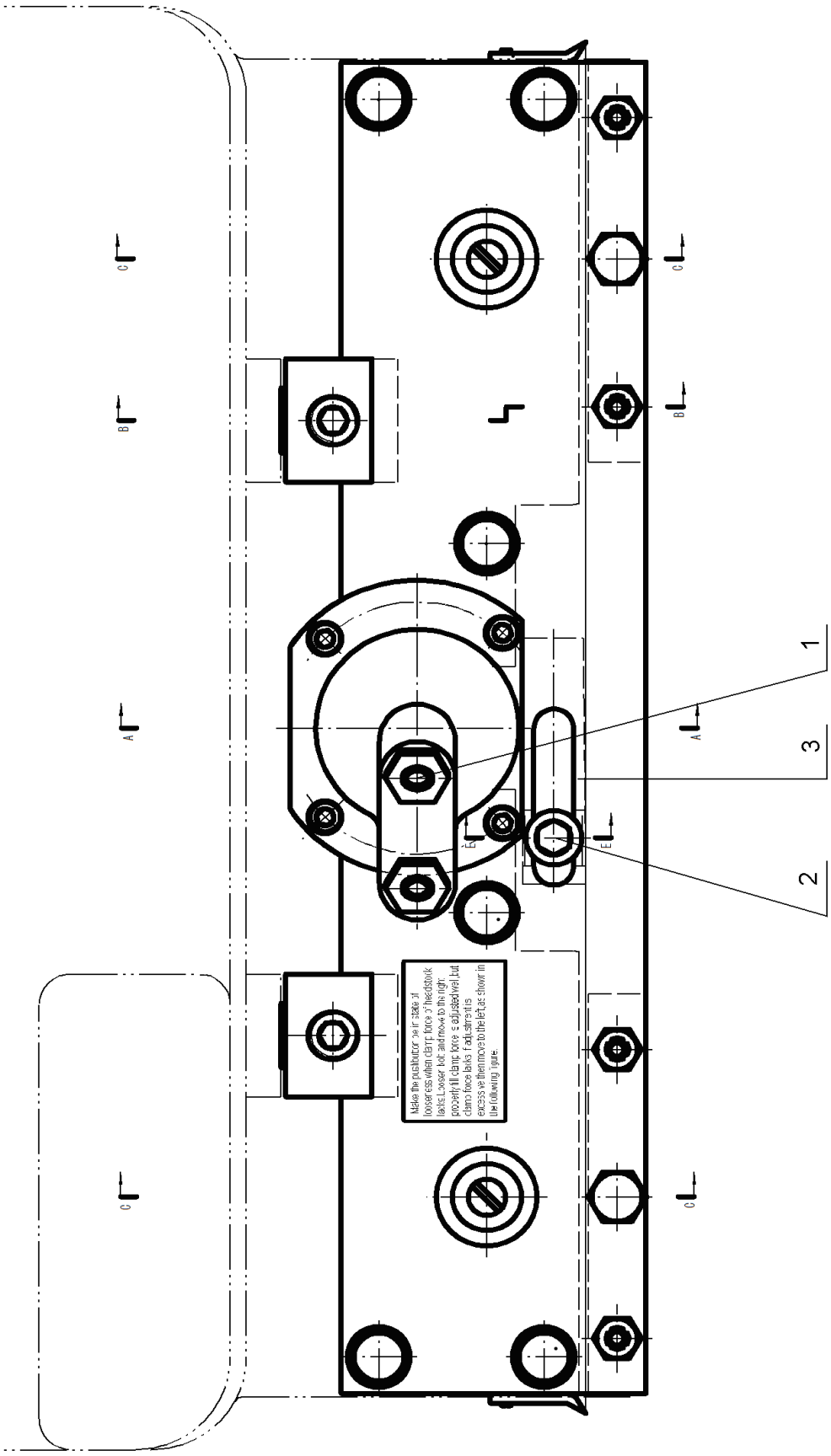

1
2
3
Make the pushbutton valve state of
looser ess when clamp force of headstock
lacks L oose bolt and move to the right
properly till clamp force is adjusted well, but
clamp force lacks f adjustments
excess ve then move to the left as shown in
the following figure.

Figure 2.3

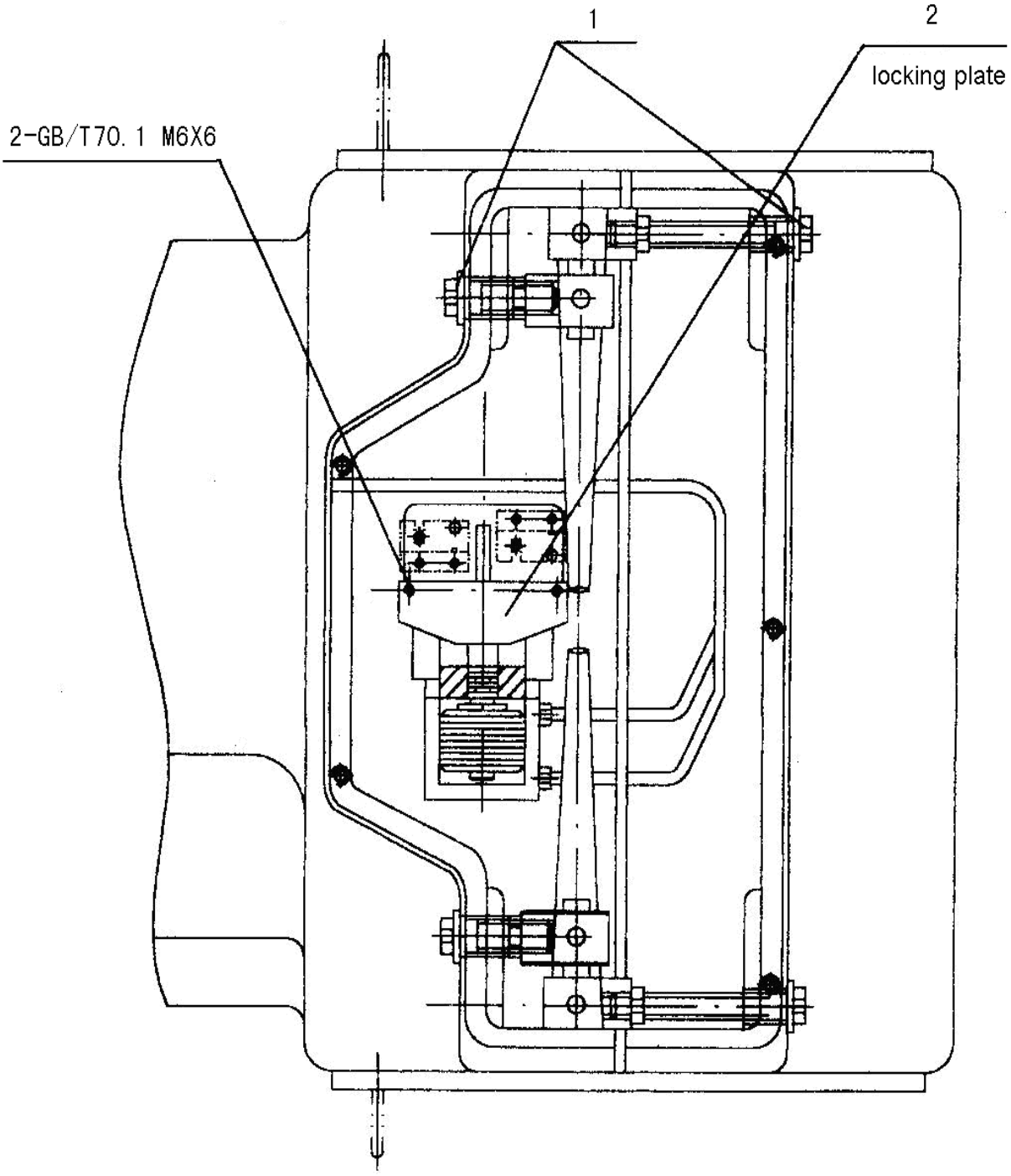
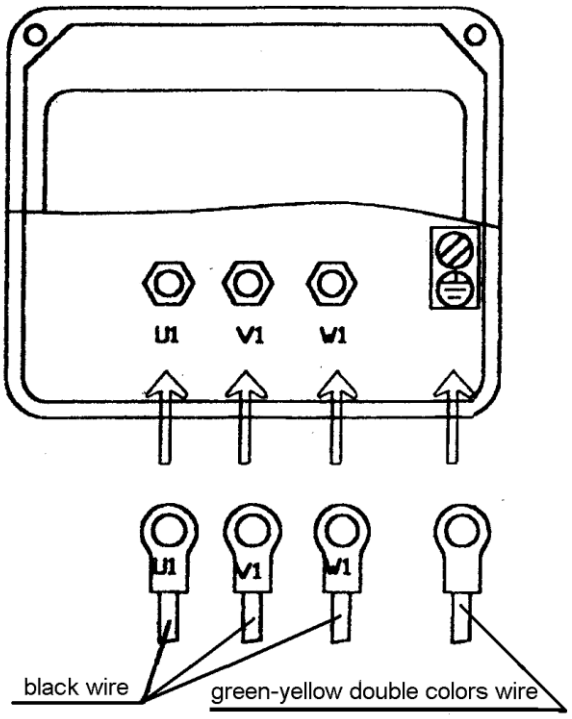
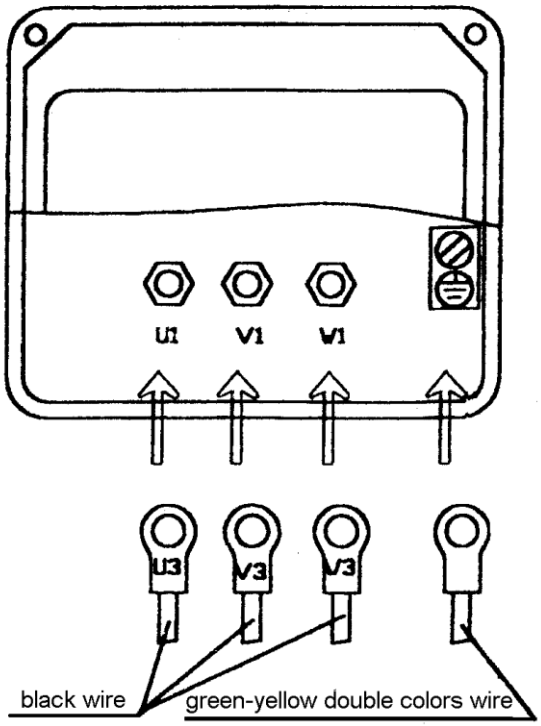


Figure 2.4

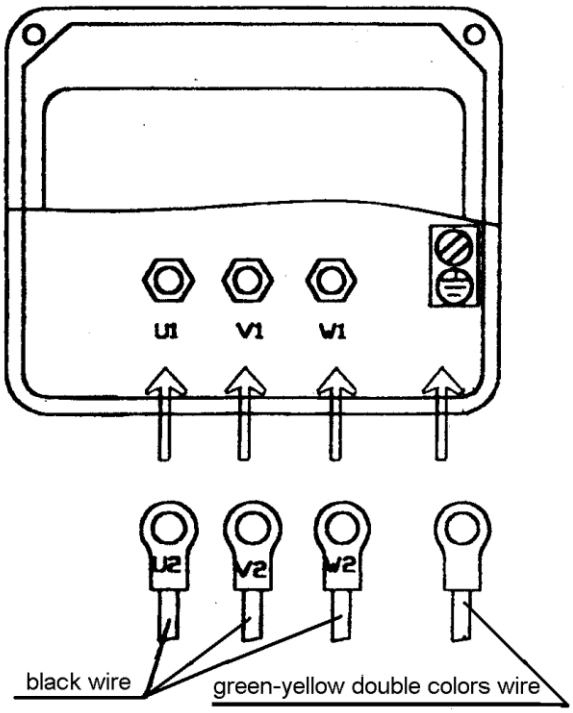
M1 spindle motor wiring box



M3 hydraulic motor wiring box



M2 elevating motor wiring box



M4 cooling pump motor wiring box

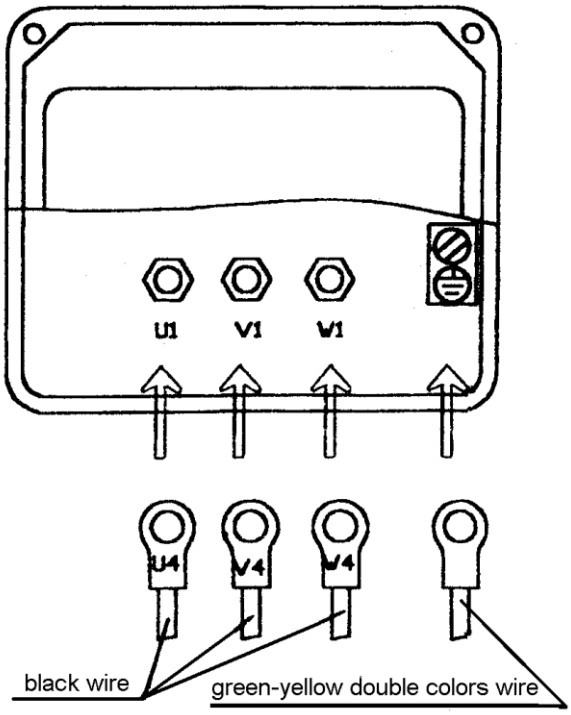
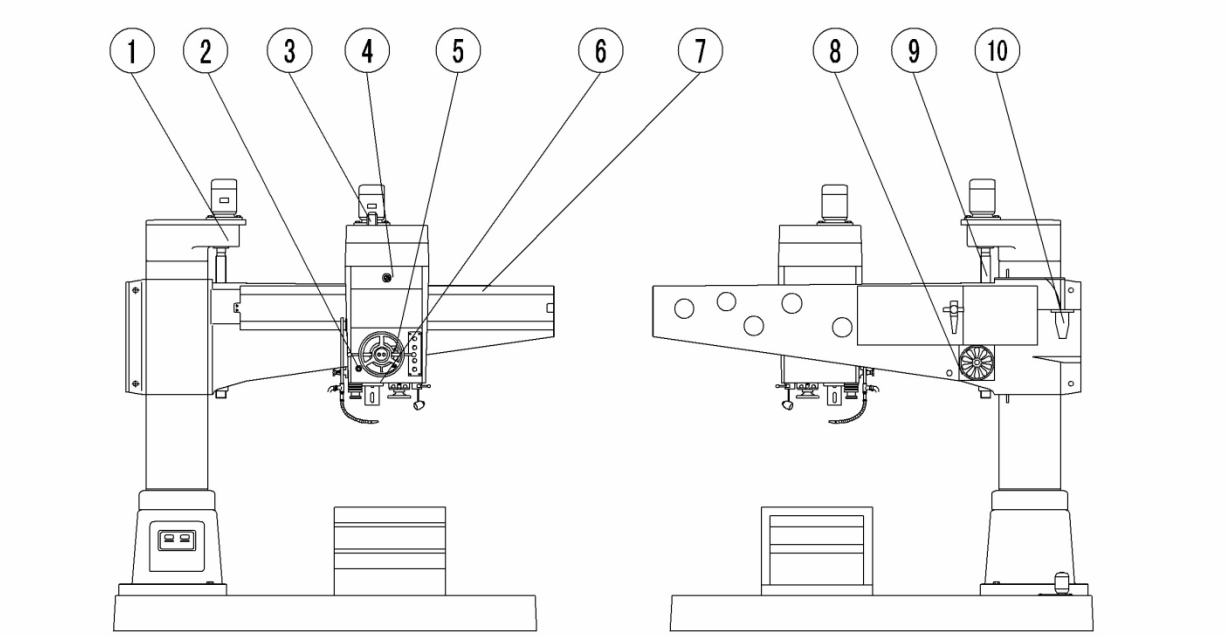


Figure 2.5



No.	Oil site	Lubricant	Cycle	Remark
1	rockerarm lifter oil basin	L-AN32	Change oil once three months.	Oil after unscrewing oil plug.
2	oil basin under gear-box	L-AN32	Change oil once three months.	Oil after opening scutcheon of gear-box.
3	spindle spline	L-AN32	Add oil once a week.	Do not oil too much.
4	oil basin on gear-box	L-AN32	Change oil once three months.	
5	inching worm	L-AN32	Add oil once a week.	Do not oil too much.
6	bearings on spindle	No.2 albany grease	Add oil once a month.	Oil after opening scutcheon of gear-box.
7	rockerarm guide	L-AN68	Six times per shift.	
8	clamped oil pump oil basin	L-AN32	Change oil once three months.	Oil after opening electrical door.
9	rockerarm lifter lead screw	L-AN68	Add oil once a week.	Do not oil too much.
10	column lubricant pump oil basin	L-AN32	Six times per shift.	

Figure 2.6

Remark: L-AN32 is the same as mechanical oil 20#

L-AN68 is the same as mechanical oil 40#

2.4 Please read the installation manual carefully after you open the package, if you choose the machine which is disjointed packing type, install them as the required sequence.

2.4.1 Clean the dust and the antirust on the surface of each part of the machine.

2.4.2 Swing the base according to picture 2.8, put the sizing block 1-4 and base as showed in picture 2.1, screw the nut to avoid the dumpage after the arm covered the column.

2.4.3 Swing the column according to picture 2.9, 2.10, clean the commissure surface, then unpack the cover on the column, put the column in the base, fix the bolt.

2.4.4 Install the arm on the column, tighten the rigging well as the picture 2.11. Make sure the arm is in balance state when it is lifted. check and clean the dust before the arm is inserted the outside column, clean the column surface, wipe the lubrication oil on the surface when the arm is inserted in the column, correct the key on arm and key groove in the column, then place arm slowly into the column. Please lay the sizing block 50—100mm under the flange which is at the bottom of rocker after the arm is inserted in the column, then unlade the swinging rings and stop up the screw at last.

Attention: the arm is not in a safe state, be care with the arm, avoid running!

2.4.5 According to picture 2.12, install the spare parts on the top of column: Remove the cover 4 of the rind and put the rind on the column, tighten the bolt 10; revolve the thread screw into the rind by hands, place the gear 11, joint the gear of motor well, tighten the nut 12, loosen the bolt which is used to fix the rhombic block on the clamp lever, connect the clamp oil pipe.

2.4.6 Remove the locking bolt (1) as showed in picture 2.3, joint the clamp oil pipe.

2.4.7 Electric installation:

2.4.7.1 According to picture 2-13, infix the power source leads through the top of the column, joint the lifting motor. (see picture 2.3.6.1)

2.4.7.2 Hydraulic pressure motor installation (see picture 2.5): Install the motor in a correct way and tighten the bolt, open the junction box and connect the leads according to the NO. on the cover and tighten it.

2.4.7.3 Main motor installation (see picture 2.5): Install the motor in a correct way and tighten the bolt, open up the junction box and connect the leads according to the NO. on the cover and tighten it.

2.4.7.4 Coolant pump installation (picture 2.5): after installing the motor and tightening the bolt, open the junction box and connect the leads according to No., cover and tighten it.

2.4.7.5 Connect the leads of main power.

2.4.8 Check the correctness of power supply phase sequence (picture 3.1)：push the button 1-18, check whether the headstock is loose (or push button 1-19, check the headstock and see whether it is clamped); push elevating button 1-3 on the arm，move the arm up by 10mm，check whether it is lifting or not (or push the down button 1-2，check whether it is lowering).

2.4.9 Push the falling button 1-2(picture 3.1)，make the arm come down 10mm，clean the antirust oil on the top of column, move the arm up 10mm again, clean the antirust at the bottom of the column, repeat the work until the oil is rubbed up. Push the button to loosen the headstock, turn the moving handwheel 1-10, move the headstock to the left side, clean the antirust oil on the guideway, then reset and clamp the headstock.

2.4.10 Trickle the lubricating oil on the surface of the column, then move the arm 50mm down, clean and lubricate the bare section again, move the arm 100mm up, clean and lubricate the bare section adequately, so operator can lift and lower the arm in a large extend.

Attention: Do not scratch the surface of the column, oiling them according to picture 2.6 and the lubricate list and check the lubricating situation, and immit the eligible coolant.

2.4.11 Move the arm up to the top of the column, then move down, in order to check if the travel switch is good.

2.4.12 Adjust level carefully, Adjust the sizing block according to picture 2.1(1,2,3,4),check the level of the cross and longitudinal of the base and make sure the level of the cross and longitudinal of the base is under 0.04/1000.

2.4.13 Turn on the current switch, adjust clamp force between the column and the

rocker, make sure the clearance between the bore of arm and column is no more than 0.04mm after clamp.

2.4.14 Adjust the precision of each item, make sure the measure values do not exceed the values on the accuracy test list, Fill the foundation bolt with cement and sand slurry after the adjustment, fix the sizing block and machine base with cement after it is vulcanized completely in order to avoid the change of the level, operator should check the accuracy with the gradient.

2.4.15 Assemble the dustproof rings which were taken-down when operator packed the machine.

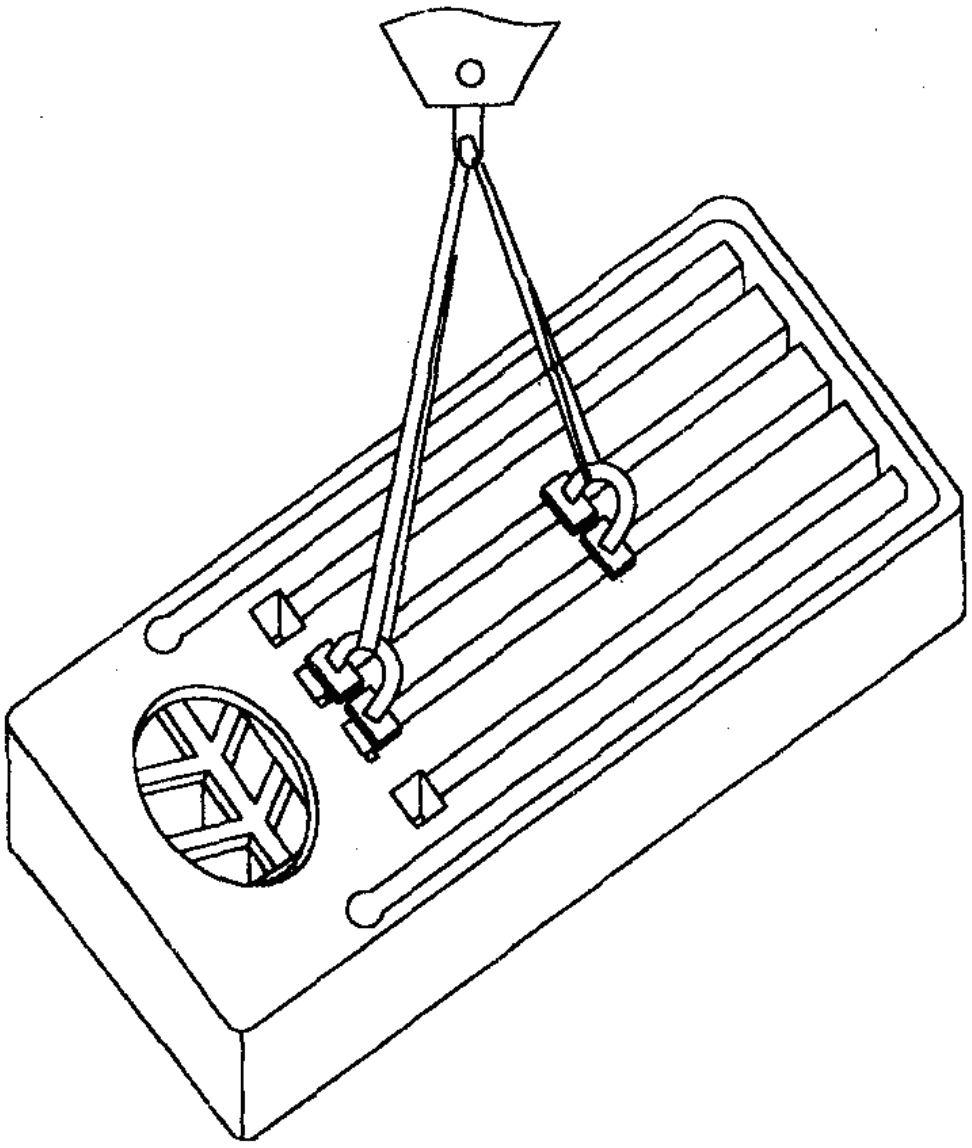


Figure 2.8

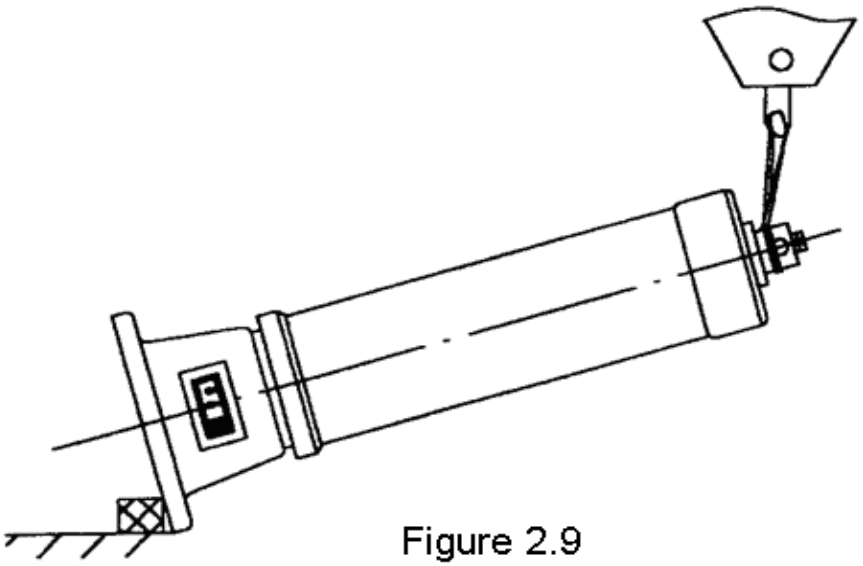


Figure 2.9

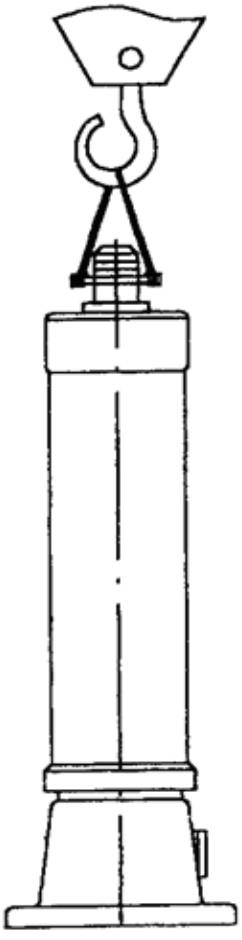


Figure 2.10

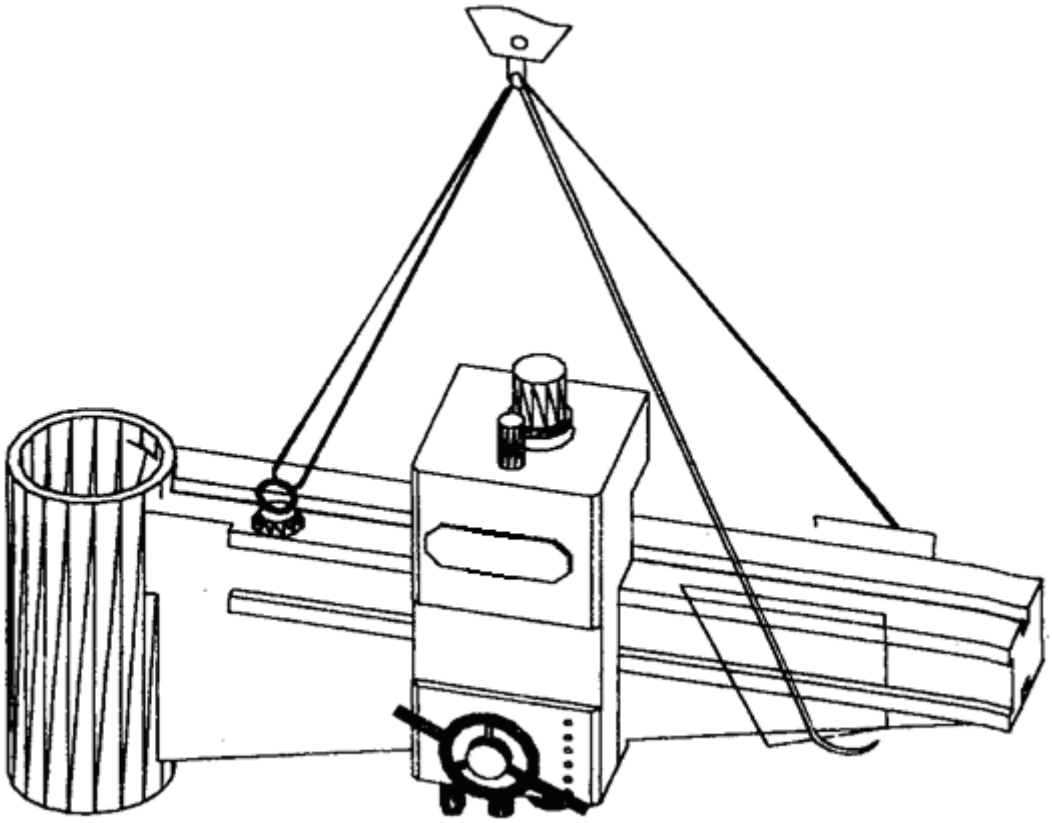


Figure 2.11

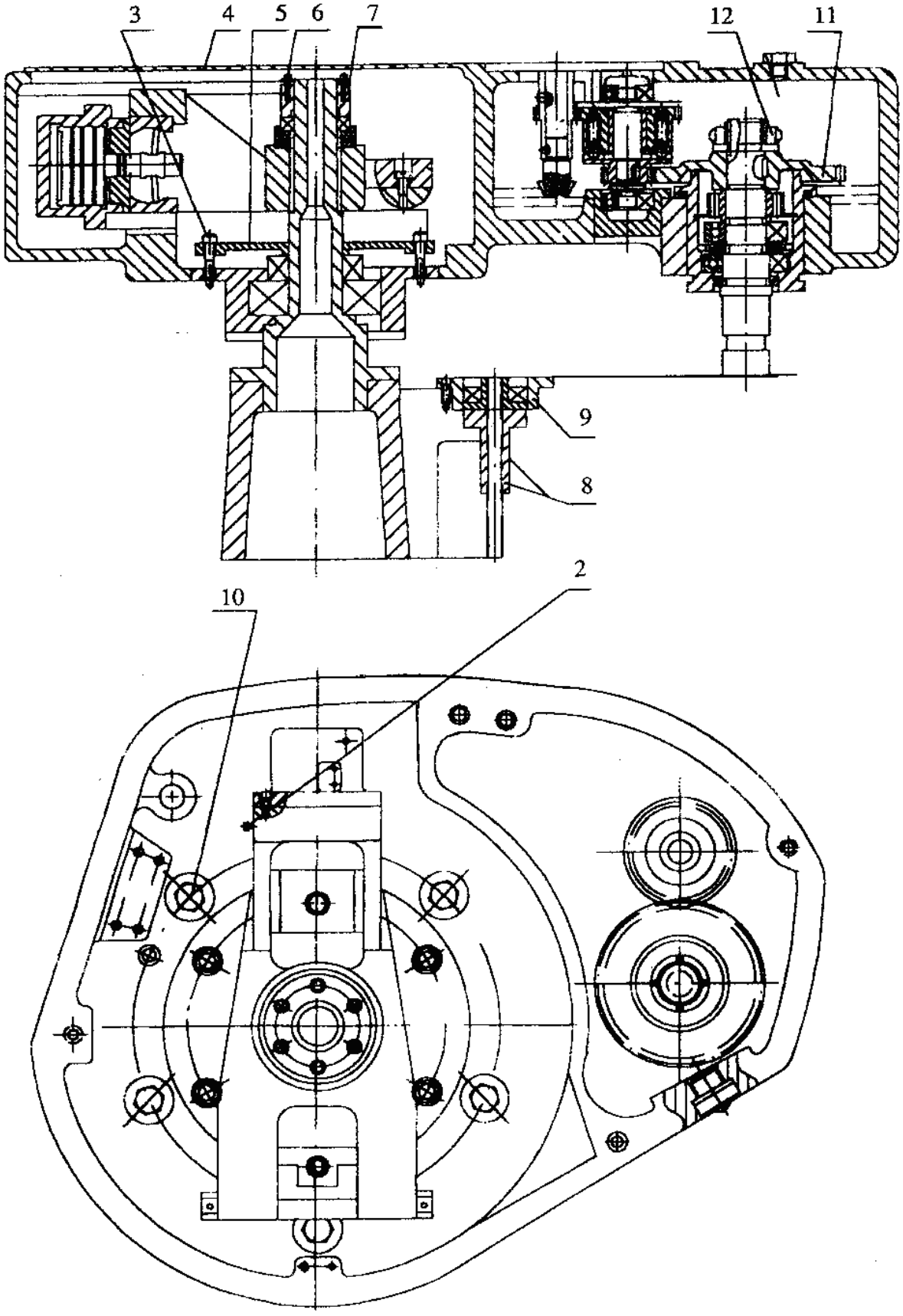


Figure 2.12

2.5 Trial run

2.5.1 Check the phase and power switch after installing the machine .

2.5.2 Start the machine (refer to the operating instruction section) and check every part is normal, i.e. the clockwise and counterclockwise turn of spindle, spindle speed changes and neutral, spindle speed of all steps, spindle feed of all steps, start and stop of spindle, elevation and fall of arm, the clamp or looseness of headstock and column, the maximum stroke of headstock, the maximum travel of arm, the manual and power feed of spindle.

2.5.3 If the above all terms are normal, then start the machine and run idle for more than 30 minutes. If everything is ok, the machine can be used.

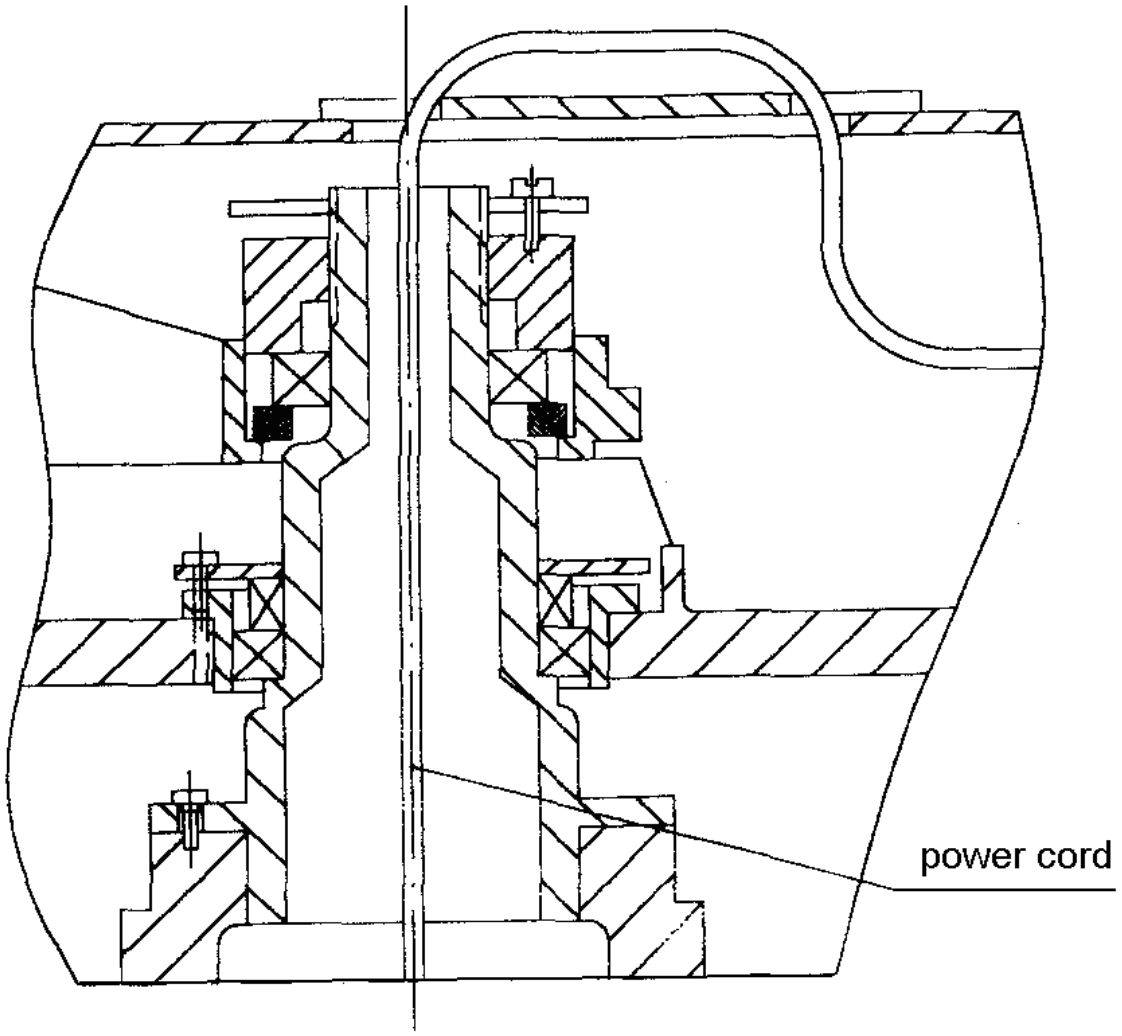


Figure 2.13

3. Technical characteristic

3.1 Main function:

The machine is widely used to drill, enlarge holes, ream, countersinking and tapping on medium-large size cast iron and steel.

Warning: Do not process flammable and explosive metal, e.g. pure aluminium and magnesium, etc.

The main handles and pushbuttons are located on the headstock.

3.1.1 External column turns around the internal column ±180° at most.

3.1.2 Arm moves up along the external column.

3.1.3 Move headstock along arm.

3.1.4 Headstock (see Figure 3.1)

Function	Handle or pushbutton
CW, CCW and neutral of spindle	1—13
manual / power feed of spindle	1—15
Horizontal movement of headstock	1—10
Spindle movement	1—9
Spindle setting stroke cutting	1—7
Spindle micro feed	1—17

3.1.5 Lift and clamp of arm: Lift power of arm is supplied by elevating motor, the drive of lead screw and nut to realize the lift of arm. (see Figure 3.1 headstock panel)。There is a safety nut on the elevating nut, which ensure the arm cannot fall suddenly. Hydraulic pressure drives rhombic block to clamp arm. Rhombic block will self-lock after clamping. Arm automatically clamp after arm lifting or lowering action finishes, which is controlled by electric switch on the oil cylinder.

Function	Handle or pushbutton
Arm up	1—3
Arm down	1—2

3.1.6 Clamp of headstock and column：Hydraulic pressure drives rhombic block

to clamp headstock and column. The two clamp actions can carry through at the same time; The two clamp actions can carry through separately too.(pushbutton 1-6 on the panel of headstock), middle position is tight, left position is to loosen the column separately, right position is to loosen the headstock separately. (pushbutton 1—18，1—19) (see Figure 3.1)

Handle list (see Figure 3.1)

Position	Name	Position	Name
1—1	E-stop pushbutton	1—11	Preset spindle feed knob
1—2	Arm down pushbutton	1—12	Preset spindle speed knob
1—3	Arm up pushbutton	1—13	Spindle handle of clockwise, counterclockwise, neutral
1—4	Main motor stop pushbutton		
1—5	Main motor start pushbutton	1—15	manual/power feed switch handle
1—6	clamp selected knob of Headstock and column	1—16	switch
1—7	The handle of setting travel and limit position	1—17	Micro feed handwheel
1—8	Dial inching adjusting handle	1—18	loose pushbutton of headstock and column
1—9	Spindle moving handle	1—19	clamp pushbutton of headstock and column
1—10	Headstock moving handle	l—20	coolant adjustable valve knob

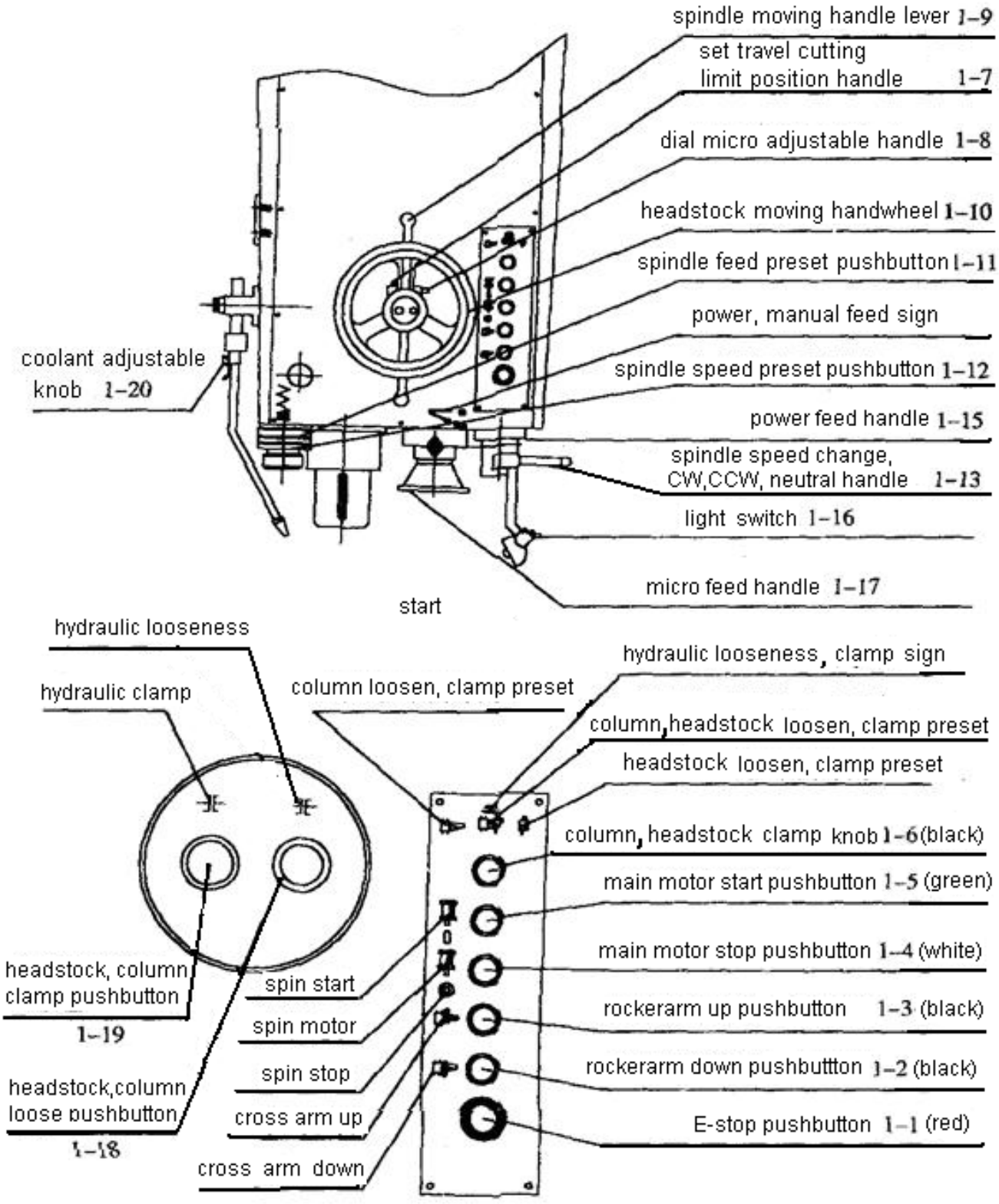

spindle moving handle lever 1–9
set travel cutting limit position handle 1–7
dial micro adjustable handle 1–8
headstock moving handwheel 1–10
spindle feed preset pushbutton 1–11
power, manual feed sign
coolant adjustable knob 1–20
spindle speed preset pushbutton 1–12
power feed handle 1–15
spindle speed change, CW,CCW, neutral handle 1–13
light switch 1–16
micro feed handle 1–17
start
hydraulic looseness
hydraulic clamp
column loosen, clamp preset
hydraulic looseness, clamp sign
column,headstock loosen, clamp preset
headstock loosen, clamp preset
column, headstock clamp knob 1–6(black)
main motor start pushbutton 1–5 (green)
main motor stop pushbutton 1–4 (white)
rockerarm up pushbutton 1–3 (black)
rockerarm down pushbuttton 1–2 (black)
E-stop pushbutton 1–1 (red)
headstock, column clamp pushbutton 1–19
headstock,column loose pushbutton 1–18
spin start
spin motor
spin stop
cross arm up
cross arm down

Figure 3.1

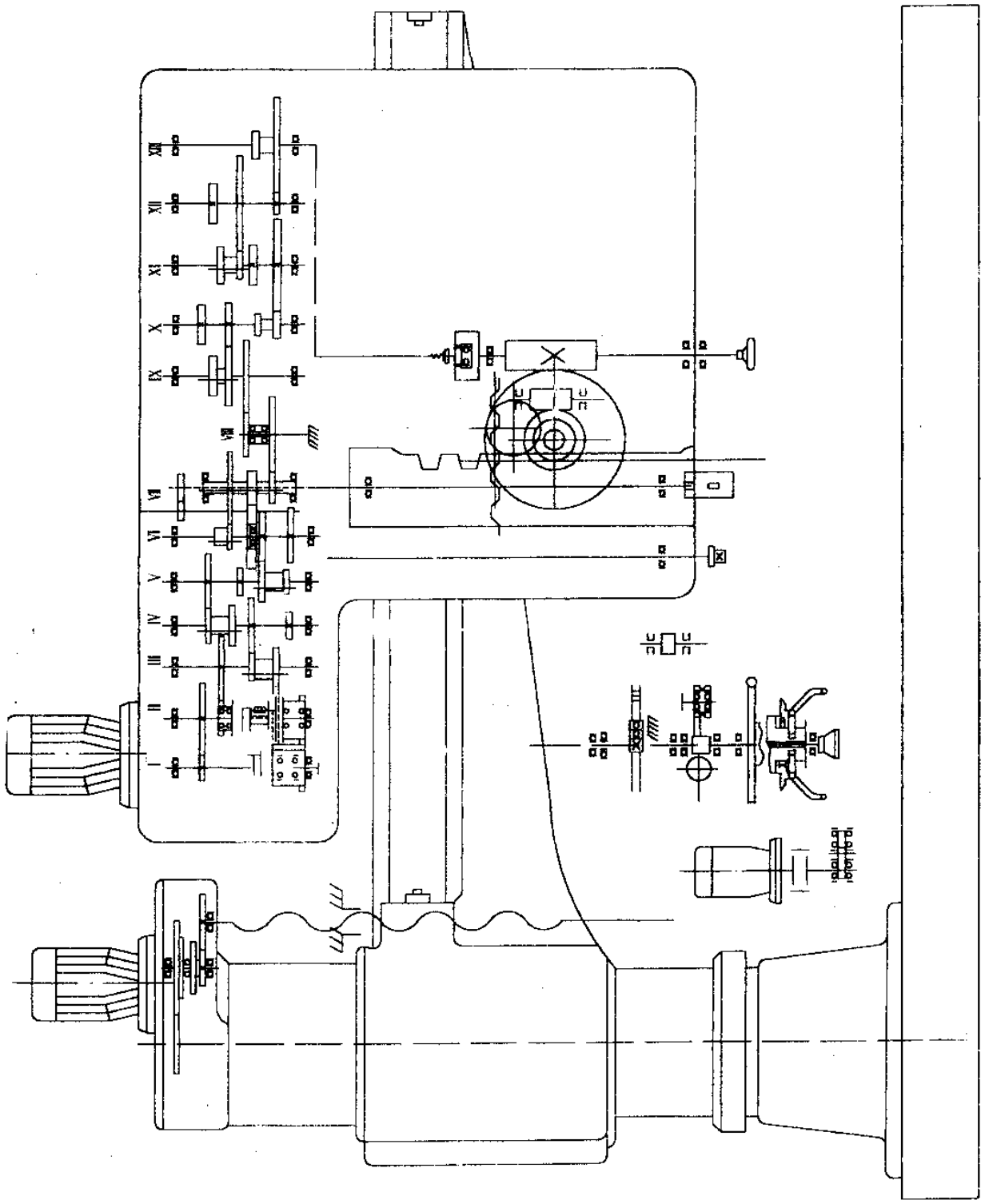


Figure 3.4

4. Operation of the machine

Operate the machine only after trial run.

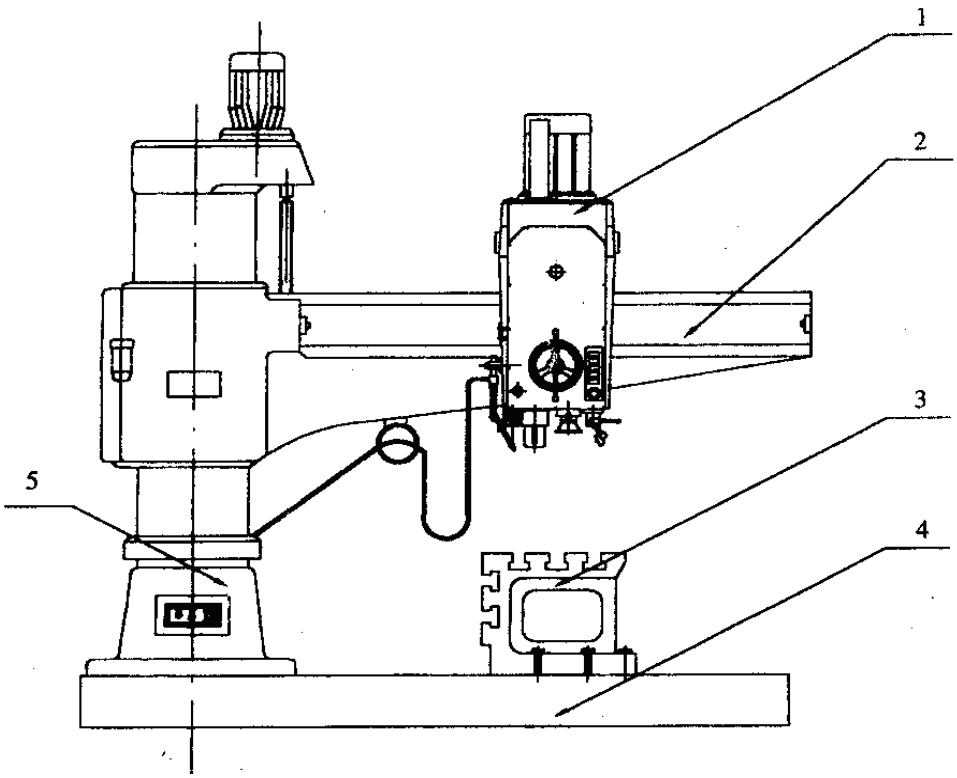


Figure 4.1

No.	1	2	3	4	5
Name	headstock	arm	table	base	column

HINT：1．Check whether locking plate in Figure 2.4 is removed;

2．Check whether joint is removed and oil pipe is connected well. (1 in Figure .2.3).

4. 1 Preparation steps as following:

4.1.1 Open the main switch 4-1 on the column, then push the startup button of coolant pump 4-3(FIG. 4.4)，and switch on the switch of the lamp (FIG.3.1);

4.1.2 Clamp the workpiece on the worktable.

4.1.3 Adjust the arm to the appropriate position according to the exact height of the workpiece. (button 1-2,1-3) (FIG.3.1, the following to 4.8 is the same)

4.1.4 Circumgyrate the knob 1-6 to choose the clamp mode of the headstock and

column (take linkage clamp as an example).

4.1.5 Loosen the clamp of the headstock and column through pushing the button 1-18;

4.1.6 Adjust the headstock to the appropriate place along the guide way of the arm through turning the handwheel 1-10;

4.1.7 Adjust the arm horizontally to the appropriate place through turning the handwheel 1-10；

4.1.8 Push the button 1-19 and make the headstock and column clamped synchronously.

4.2 Automatically drill：

4.2.1 Push the startup button 1-5 of the main motor；

4.2.2 Circumgyrate primary knob 1-12 of the spindle speed and choose appropriate speed：

4.2.3 Circumgyrate primary knob 1-12 of the spindle feed and choose the appropriate feed；

4.2.4 Use setting stroke cutting limited handle (see the detailed spindle feed later)；

4.2.5 Push down the automatic feed handle 1-15, and make it link to the feed state；

4.2.6 Pull out the handle 1-9, put through the automatic feed；

4.2.7 Speed change of the spindle: Push down the CW, CCW handle 1—13 to the position of speed change gears for 3 seconds, achieve the speed of primary election and feed; then put up in horizontal position, and move the handle left to the clockwise position, then automatic drilling and cutting can be carried through.

4.3 Two instances of manual feed:

4.3.1 Change automatic feed into manual feed: During the above process automatic feed, if change it into manual feed, just need to push inside the handle 1-9, disconnect automatic feed, and then use the hand to control the handle 1-9；

4.3.2 The manual feed: when operation begins, if not push down the handle 1-15 of automatic feed, automatic feed cannot be performed; contra-rotate the handwheel 1-9 to realize manual feed.

4.4 Tapping

4.4.1 Push the startup button 1-5 of main motor；

4.4.2Circumgyrate primary knob 1-12 of the spindle speed and feed knob 1-11, choose spindle speed and feed rate.

4.4.3 Circumgyrate the spindle clockwise or counterclockwise, push down the speed change handle 1-13 to speed change position for three seconds, put it up to the position of clockwise rotation, the screw thread can be drilled, after reaching preset depth, move the handle 1-13 to the position of reversal to unthread the tool.

4.5 Explanation of spindle neutral /speed change：

Do as fig.4.2, put up the handle 1-13, it is easy to turn the spindle by hands, if want to start up spindle, push down 1-13 to the position of speed change until the spindle turns, and then move it to clockwise rotation and reversal.

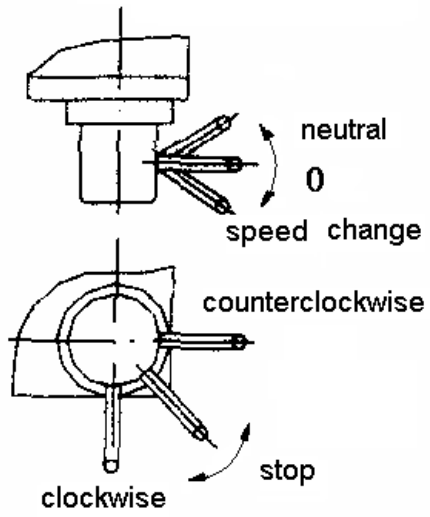


Figure 4.2

4.6 Explanation of the spindle clockwise and counterclockwise rotation

As shown in the fig.4.2, adjust the handle to the position of the clockwise and counterclockwise rotation after the main motor starts up, which means the spindle turns clockwise and counterclockwise.

4.7 Explanation of the spindle speed

4.7.1 Automatic feed：Push down the handle 1-15, then pull out the handle 1-9, then automatic feed is to carry through。

4.7.2 Manual feed: Push the handle 1-9, rotate it, drive the spindle feed up or down.

4.7.3 Micro feed：Put the handle 1-15 in the horizontal position and then pull out handle 1-9, turn handlewheel 1-17.

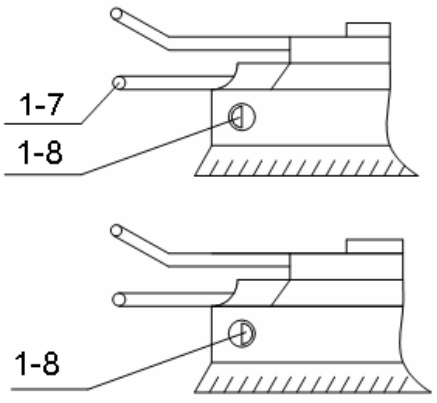


Figure 4.3

4.7.4 Setting stroke cutting：

As shown in the fig. 4.3, pull out the handle 1-7, and after turning the handle 1-8 to the position in the picture, turning the dial to the needed cutting depth, and align with “0” of box size, and then turn handle 1-8 to the position in the fig.4.3, adjusting slightly until

align with “0”, and clamp handle 1-8 using clamp knob on the other end, push handle 1-7, switch on feed switch, when cutting depth reaches preset stroke value，handle 1-15 is put up automatically and complete cutting.

4.8 Operated procedure of stopping the machine：

4.8.1 Put up automatic feed handle 1-15, disconnect automatic feed.

4.8.2 Push moving handle 1-9 of the spindle, disconnect automatic feed.

4.8.3 Clamp headstock and column.

4.8.4 Stop the running of main motor by pushing the button.

4.8.5 Turn off the coolant switch.

4.8.6 Turn off the switch of the main power supply.

ATTENTION：

1. When drilling to the critical state of demand size， feed capacity must be minimised to avoid destroying the tool.

2. If operating wrongly, the worker must push the red emergency-stop button (1-1) on the plate of the headstock at once.

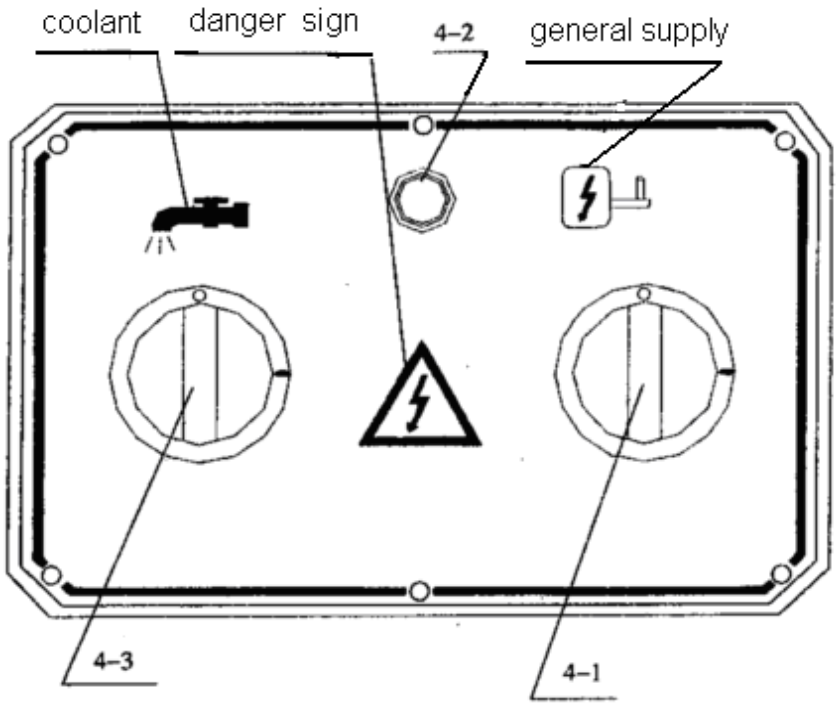


Figure 4.4

Name of switch

No.	Name
4—1	Main switch(red)
4—2	Power light
4—3	Cooling switch(black)

5. Troubleshooting

Symptom	Possible cause	Corrective action	Remark
Main motor does not turn when pushing the startup button.	Main switch is not switched on；startup button does not connected well; contactor does not attract; junction box of the main motor and connected point are not connected well or joint break off.	Open the main switch, check the joint of the startup button; check every connected point of contactor in the junction box, and connect them. Replace bad one with new one if necessary.	
spindle speed change operation handle has fault	Speed change oil pump is single direction, the motor's turning is wrong, the pump does not inhale the oil, so it cannot drive the shifting fork to push friction disk up or down	Change random two wires of the main motor.	
The spindle and surface of the machine leak oil.	There is too much oil in the oil box, it leaks from the top of the bearing house.	Do not exceed the mark when adding oil.	
Lead screw of box leaks oil.	There are too much oil in the oil box,, the redundant oil leak out from the bearing on the screw.	Oil according to the mark, do not exceed the mark.	
The headstock cannot be clamped.	The clearance of 55° guide way between headstock and rocker is too big; the position of clamp block 3 is wrong; the connection between the clamp oil cylinder and brake frame leaks oil or oil pipe 1 leaks oil; magnitude of interference of rhombic block is not enough, it cannot self-lock.	Adjust bolt 5 to make the clearance between the 55° guide ways not more than 0.04mm,(check with the 0.04 clearance gauge); in the state of looseness, loosen bolt 2 and adjust clamp block 3 to appropriate position, then tighten bolt 2 again.	see Fig.7.1

Headstock move inflexibly or does not move after pushing the pushbutton of headstock.	1. Loosen bolt 2, clamp block 3 moves too when moving headstock, which makes the clearance between clamp block and guide way reduce, so affect the movement of the headstock. 2 If feel inflexible when moving headstock, please check if the ball bearing is broken.	Check if bolt 2 loosen on the clamp block, and replace the ball bearing on the clamp plate.	See Fig.7.1
The column cannot be clamped	Many bolts are adjusted too much on spring plate, which makes column put too high, rhombic block clamp organ cannot make the inner and outside surface of column tight; the nut 6 is not adjusted appropriately, which will produce clearance between lever and box; there is not enough pressure oil to drive the piston, so make rhombic block cannot be in the state of clamping.	Readjust the blot on the spring plate, but must make sure the force when rocker is loose, then adjust the nut on the column to make the clamp force reach 1568N(thrust in the end of rocker), then use the bolt to tighten the nut; adjust the nut 6 to diminish the clearance between lever and box if loosen the column too little, then tighten the 4 bolts in the nut 7; adjust the oil in the hydraulic system and make sure the oil is in the state of 245 ～ 291X10^4Pa or changing the oil in the oil tank.	See Fig.7.2

The arm cannot be clamped.	Bolt 1 is not adjusted appropriately, there is clearance between the arm and outside of column when rocker is in the state of clamping, the limit switch is near the piston, piston touched the limit switch when piston does not reach the position of clamping, which make hydraulic motor stop supplying the oil. 。	Remove the side cover, loosen clamping nut or adjust bolt 1, which not only make sure the 0.04mm clearance, and clearance gauge cannot insert it, but also make sure rhombic block self-lock；adjust the position of limit switch to make the normal close point of the limit switch disconnect in the state of clamping， and make rhombic block be in the position of clamping, please refer to headstock clamp organ about the rhombic block.	See Fig.7.3

Attention: 1. If there is serious malfunction that needs to open headstock, which must need the professional to repair the machine.

2.If there is some malfunction, it needs to test the machine again after eliminating the malfunction.

6. Disposal of malfunction

6. Disposal of malfunction

The machine is universal tool. During the process of workpiece, the spindle parts and tools can move up or down, or rotate. The operator must keep safe during the process of machining, once the urgent thing happens, such as engulfment, impact and brush burn, first the operator push down the red emergency-stop knob 1-1. and then eliminate the malfunction, switch on the main switch 4-1 after all the malfunction is eliminated.

7. Maintenance and adjustment of the machine

7.1 Maintenance：

7.1.1 The machine must be lubricated according to the operation manual during the

operation, the strainer must be cleaned on time and keep the oil clean.

7.1.2 The guide way of the rocker and column should keep clean, and should often lubricate them per shift avoid injury.

7.1.3 The table and base must keep clean, the scrap iron should be cleaned in time, do not clean them directly by hand to avoid injury.

7.1.4 Coolant must be changed every month, and keep clean, the impurity such as scrap iron in the coolant is deposited in the entrance of the coolant box, so scrap iron should be disposed every day.

7.1.5 The spindle must be cleaned before inserting the tools.

7.1.6 Maintenance of the electrical equipment:

The electrical equipment must keep clean, so the dust and dirt and so on must be cleaned in time. The loop cannot be cleaned by kerosene and gasoline. The motor should be checked once a year, and the bearing needs to be cleaned once a year. The oil must be changed once a year at lease. The bearing of the motor must be lubricated by grease. All contactor should be cleaned carefully. Abrasion contactor must be exchanged with a new one. Contactor do not allowed to oil. The frayed and oxygen part must renew with file slightly.

7.2 Adjustment of machine

7.2.1 Adjustment of spindle clamp force (see Figure 7.1)

Loosen screw 2 and remove screw 3 to adjust spindle clamp force. Bear 392N circle force on the verge of wheel, which just is ok when the headstock cannot be loose. Adjust the clearance between the arm and headstock by bolt 5 and 6, use the 0.04mm thickness gauge to examine, but the demand depth can’t be deeper than 20mm; and achieve demand of the headstock, headstock can be moved when bear circle force on the verge of the wheel, then screw down the bolts.

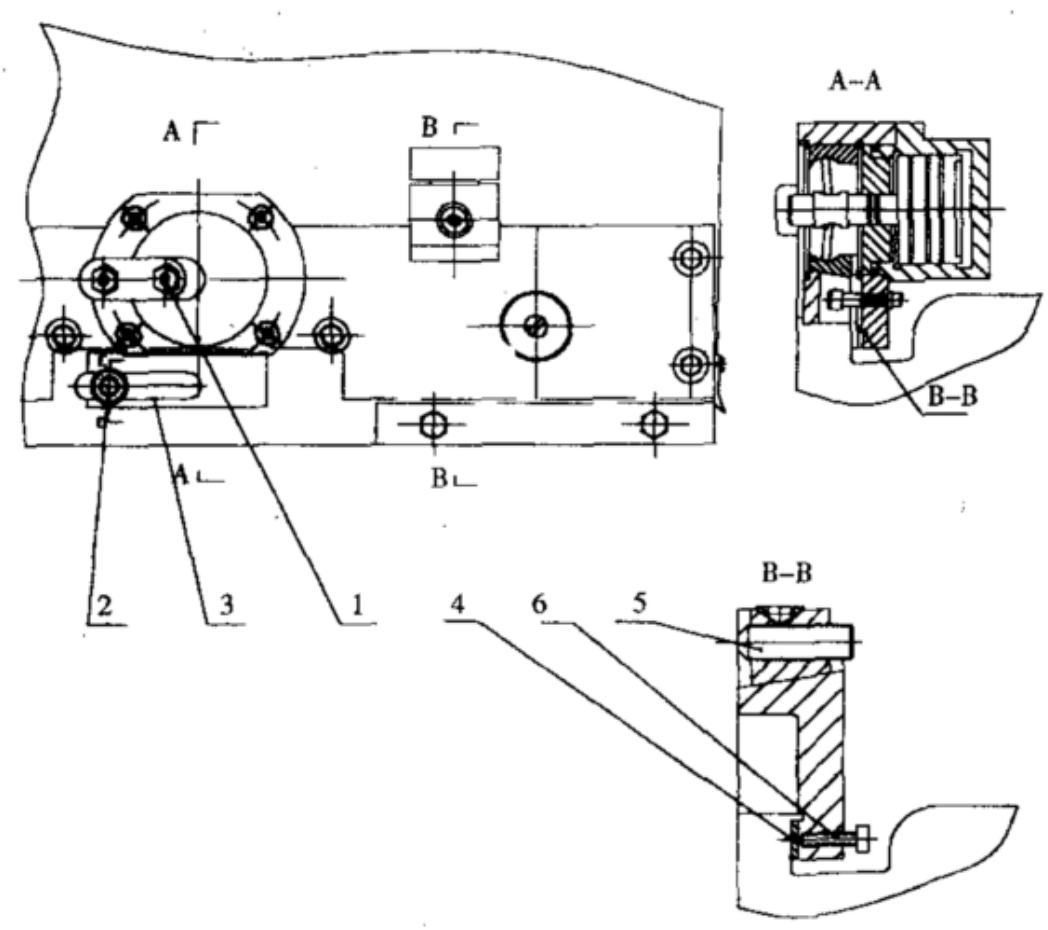


Figure 7.1

7.2.2 Adjustment of column clamp force

The bolt 3 is used for adjusting the clearance of the inside and outside cone-shaped surface of column. Please force bolt uniformly when adjusting bolt. Use the nut 7 to adjust the column's clamping force. If operator bears 1568N force on the end of arm, the column can't be swiveled, but it can swivel after loosened and bear 29N horizontal strength, please screw down bolt 6 and screw nut 7.

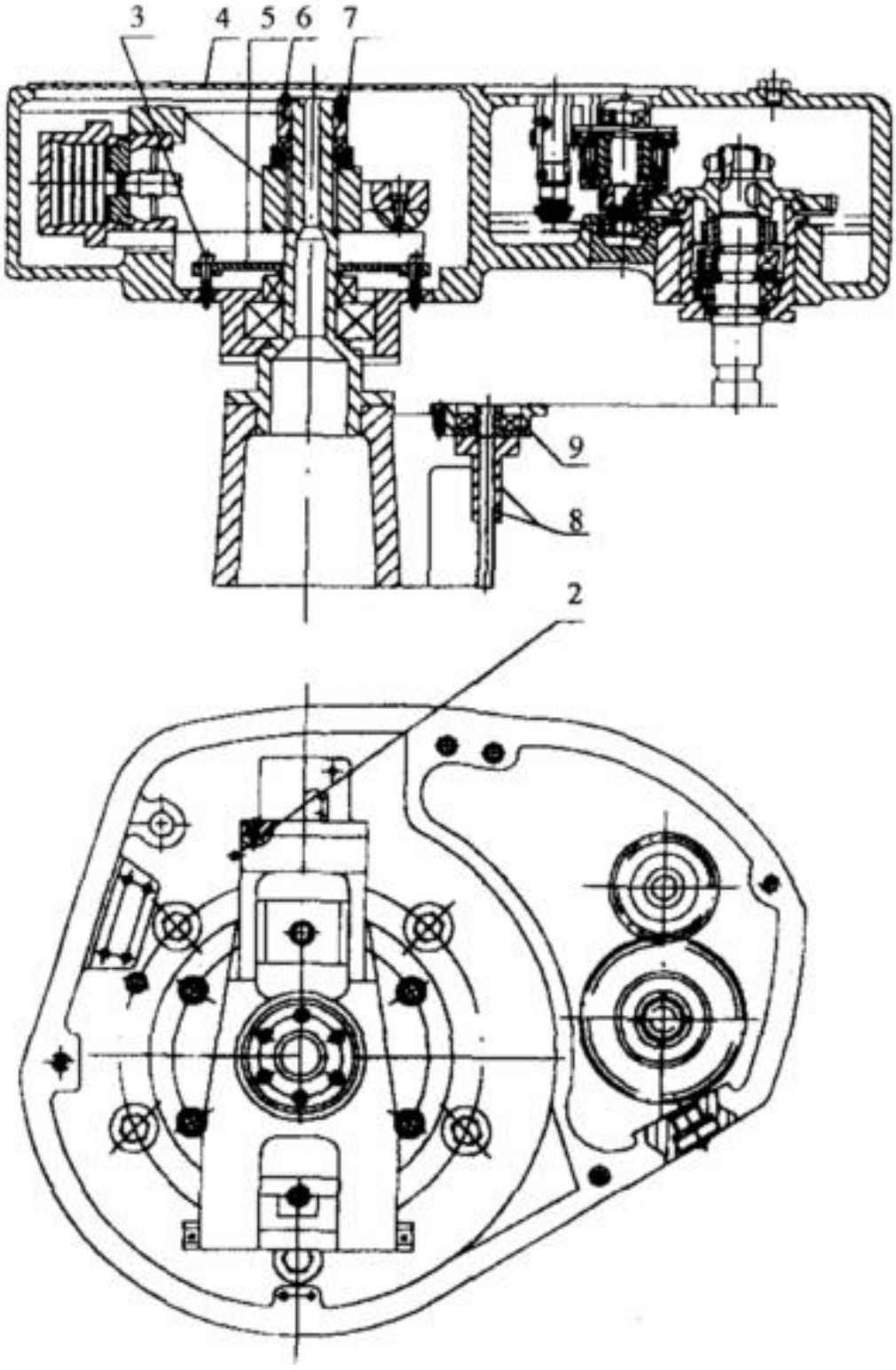


Figure 7.2

7.2.3 Adjustment of arm clamp force (see picture 7.3)

shut off the power during elevating of arm when adjusting arm, then the arm will be lax. Adjust the bolt 1, 2, 3 and 4 befittingly to make sure the 0.04mm clearance gauge can’t insert when clamping the arm.

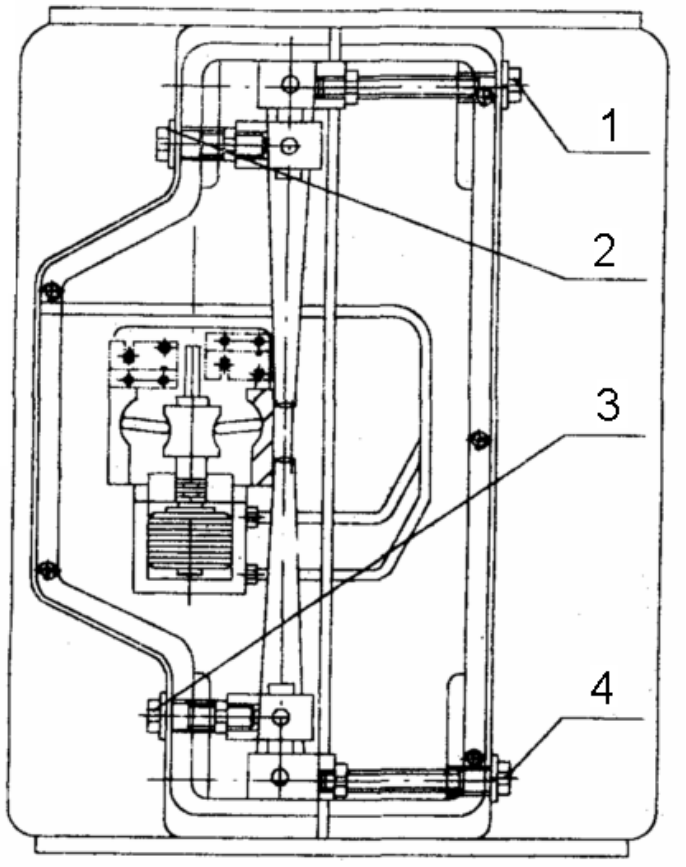


Figure 7.3

8. Hydraulic system of machine

Hydraulic system of machine consists of control organ and clamp organ.

8.1 Control organ of hydraulic system (Figure 8.2)

Control valve and preselection valve are installed at the top of headstock. Spindle speed preselection valve and feed preselection valve are six-way turning valve of 16-bit, each valve controls four differential motion oil cylinder to realize 16-step speed change. Control valve is 5-bit and six-way turning valve, which controls clockwise oil cylinder, counter-clockwise oil cylinder, speed oil cylinder, neutral oil cylinder, brake oil cylinder.

Every position of control valve, oil way state as following：

Action / No.	1	2	3	4	5	6
clockwise	+	-	-	-	+	-
counterclockwise	-	+	-	-	+	-
Speed change	+	+	+	-	+	-
neutral	-	-	-	+	+	-
stop	-	-	-	-	-	-

remark：“+”: high pressure oil， “—”: low pressure oil

connection of every oil way：

1．Clockwise oil cylinder
2．Counterclockwise oil cylinder
3．Preselection valve
4．Neutral oil cylinder
5．Oil pump outlet
6．scavenge oil mouth

High pressure oil enters preselection valve to push every speed change oil cylinder, which drives shifting gear to change speed; High pressure oil enters clockwise oil cylinder through eyelet during the speed change simultaneously; because the oil cylinder's area is different (clockwise cylinder's area is bigger than counter-clockwise cylinder's area),which causes the shifting fork push the friction disk to push clockwise friction disk a little force, then the transmission chain can be started.

When the control valve is in stop position, oil way ⑤ directly scavenge oil way ⑥,the whole system is in low-pressure state, at this time, the spring will be loose because of without the push of oil way ⑤, then it will push the friction disk

heavily to push the transmission chain start, then the spindle will stop.

Please join the oil pipe as shown in the picture 8.3, to make sure the end of No.1 oil pipe into oil net when inserting No.1 oil pipe and cleaning the oil net, or else, the hydraulic pressure system will possibly destroyed, so the oil pipe can't be used for long time.

Hydraulic pressure system's force is controlled by the spring 3 in overfalling valve on the oil pump, the force is $156.8x10^4$～$196x10^4$ Pa.

8.2 clamp organ of hydraulic system (Figure 8.1)

The oil impels the piston to realize clamp and looseness of the headstock, column and arm through rhombic block, clamp or looseness of arm is controlled by electromagnetic valve separately. But clamp and looseness of headstock and column can be done at the same time or separately. These operations are controlled by the pushbuttons 1-6 on the headstock.

When operator clamp or loosen headstock or column, please adjust the pushbuttons 1-6 to designated position (middle position is for linkage looseness, right is for loosening headstock separately, left is for loosening column separately), then to start clamp motor, the oil from the oil pump will flow into clamp oil cylinder through distributor to impels the piston and rhombic block to realize clamp.

Hydraulic system is closed circulation type: acting force is $250x10^4$～$280x10^4$ Pa. because the oil cylinder's area is different, part oil is supplied when oiling into big oil cavity, superfluous oil will leak when oiling into small cavity. Oil leaking will flow back oil tank. The system do not bring excessive pressure phenomenon and low pressure phenomenon due to low efficiency of cubage.

8.3 Coolant

Use coolant during processing the steel, ream, tapping.

Push the switch 4-3(see picture 4.4) to start coolant pump, adjust the knob 1-20 (see picture 3.1) to control flux.

Notice : Make sure the nozzle aim at the processing part.

Don't use the flammable and deleterious coolant.

model：A0B-25　　power：0.09kW

Flux：25L/min　　pump lift：4m

Coolant list：

Purpose	Component	Property	Remark
Drill steel	Emulsification plaster 3～5％，soda ash 0.2～0.35％， sodium nitrite 0.25～5%， water		
ream	Emulsification plaster 3～5％，soda ash 0.2～0.35％， sodium nitrite 0.25～5%， water	Nonflammable, antirust	
tapping	oxidation Paraffin 40%， kerosene 30％， transformer oil 30％。		Smear them on cutter before tapping

8.4 Disposal of coolant

The coolant box is at the back of machine base, coolant can be used circularly. Please renew it in time when it is dirty.

Method: open the coolant pump, then let the coolant flow into an empty container, use the dishcloth to clean the box, then you can put new coolant.

The impurity such as scrap iron which remain on the portal of coolant box, please clean it in time.

headstock clamp organ column clamp organ arm clamp organ

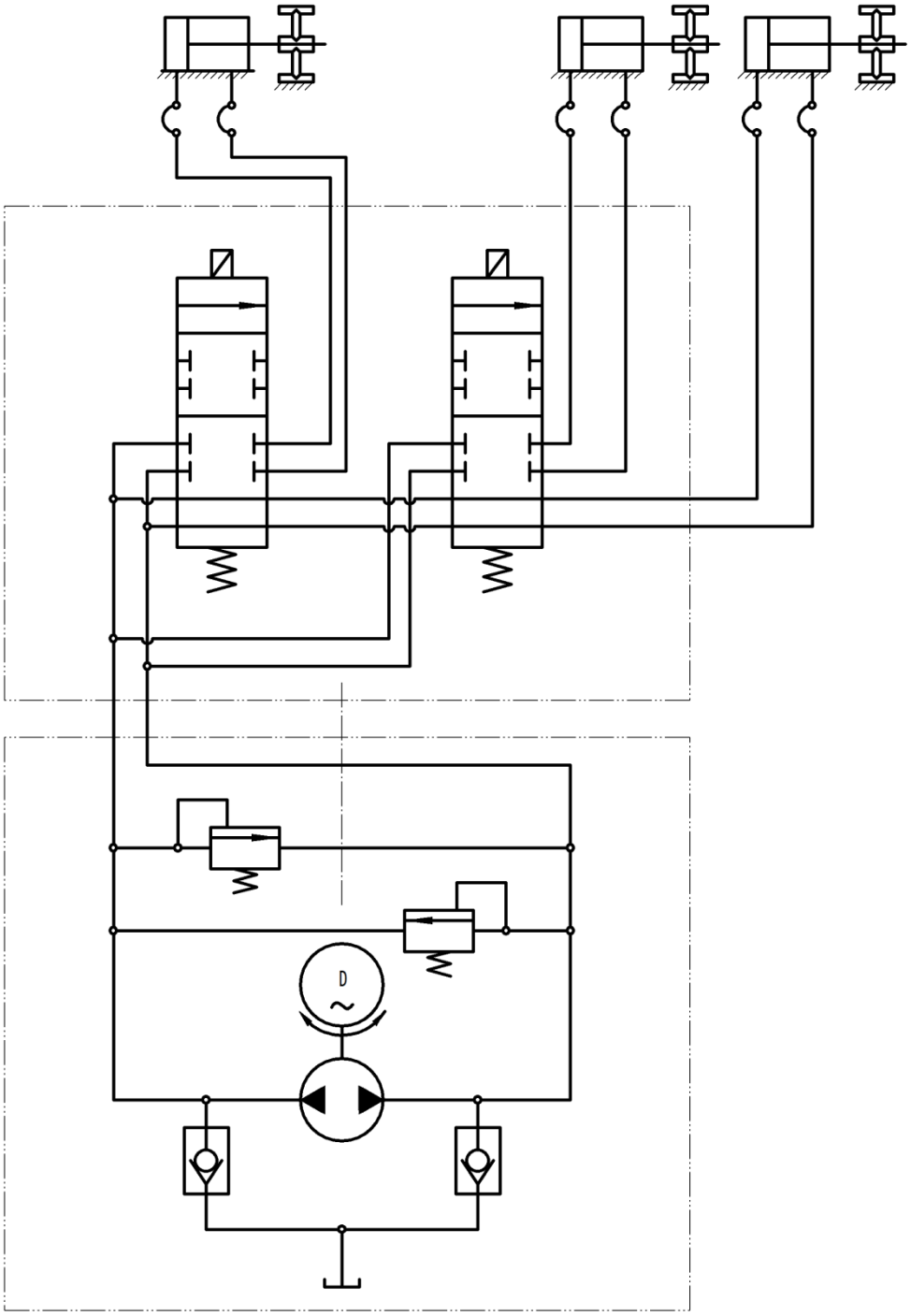


Figure 8.1

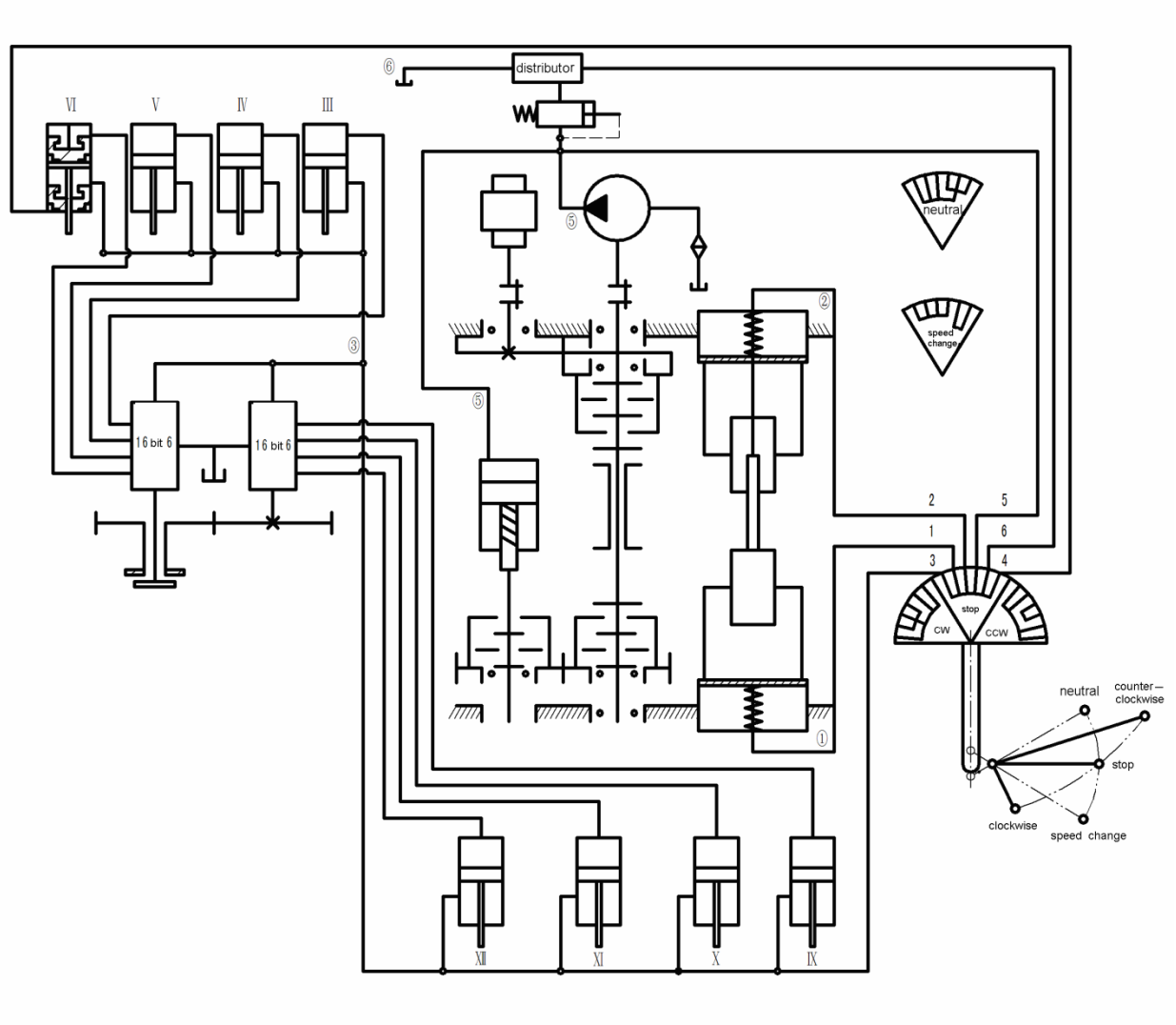

distributor
VI
V
IV
III
neutral
speed change
16 bit 6
16 bit 6
2
5
1
6
3
4
stop
CW
CCW
neutral
counter–clockwise
stop
clockwise
speed change
XII
XI
X
IX

Figure 8.2

Commutation table of speed change oil cylinder

rotating speed \ oil cylinder number	Ⅲ	Ⅳ	Ⅴ	Ⅵ	rotating speed \ oil cylinder number	Ⅸ	Ⅹ	Ⅺ	Ⅻ
2000	+	+	-	+	3.20(0.126″)	+	+	-	-
1250	-	+	-	+	2.0(0.0787″)	+	-	-	-
800	+	+	+	+	1.25(0.0492″)	-	+	-	-
630	+	-	-	+	1.0(0.0394″)	+	+	+	-
500	-	+	+	+	0.8(0.0315″)	-	-	-	-
400	-	-	-	+	0.63(0.0248″)	+	-	+	-
320	+	+	-	-	0.5(0.0197″)	+	+	-	+
250	+	-	+	+	0.4(0.0157″)	-	+	+	-
200	-	+	-	-	0.32(0.0126″)	+	-	-	+
160	-	-	+	+	0.25(0.0098″)	-	-	+	-
125	+	+	+	-	0.2(0.0079″)	-	+	-	+
100	+	-	-	-	0.16(0.0063″)	+	+	+	+
90	-	+	+	-	0.13(0.0051″)	-	-	-	+
63	-	-	-	-	0.1(0.0039″)	+	-	+	+
40	+	-	+	-	0.06(0.0024″)	-	+	+	+
25	-	-	+	-	0.04(0.0016″)	-	-	+	+

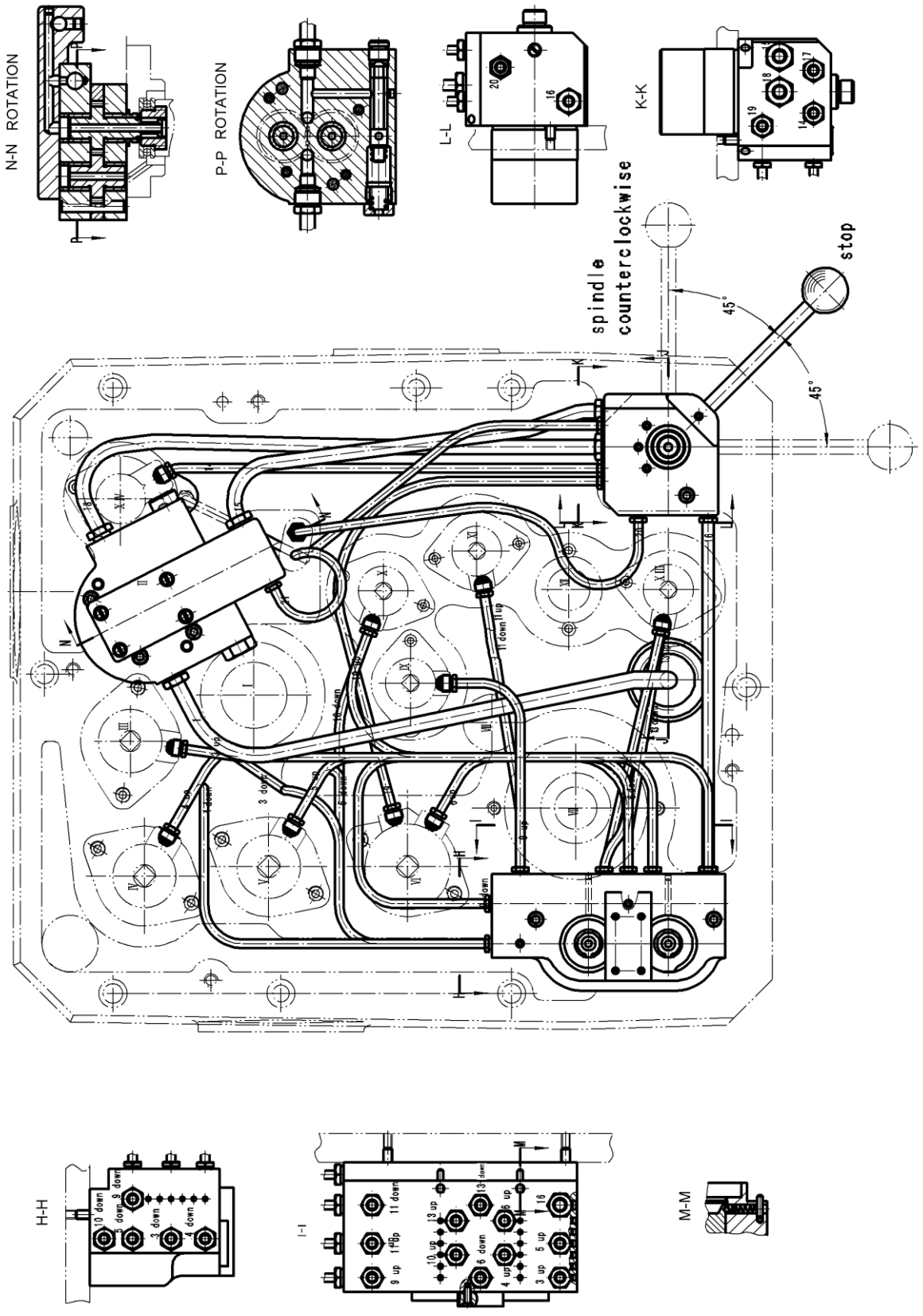

N-N ROTATION
P-P ROTATION
L-L
K-K
spindle counterclockwise
stop
45°
45°
spindle clockwise
H-H
I-I
M-M

Figure.8.3

9. Electric system of machine

9.1 Summarization

9.1.1 Rated voltage：3ph 415v。

9.1.2 Voltage range：normal voltage is 0.9~1.1 times of rated voltage.

9.1.3 Frequency：50Hz.

9.1.4 Frequency range：0.99—1.01 times of rated frequency（continuous）.

0.98—1.02 times of rated frequency（short-term operation）.

9.1.5 Harmonic wave：the summation of 2~5 times aberrance harmonic wave do not exceed 10% of line voltage square root value；the summation of 6~30 times aberrance harmonic wave at most allows 2% of affix line voltage square root value.

9.1.6 Imbalance voltage: Negative sequence and zero sequence of three-phase do not exceed 2% of positive sequence.

9.1.7 Voltage interruption: Interrupt power or zero voltage duration does not exceed 3sm, interval falling time does not exceed 1s during random time of power cycle.

9.1.8 Voltage fall: Voltage fall does not exceed 20% of peak value voltage in 1 cycle，falling interval time should be more than 1s.

9.1.9 Maximum start-up current：67.62A。

9.1.10 Control loop voltage：AC 110V，lighting voltage：AC 24V。

9.1.11 All motors which are used for transmission, as following：

M1—main motor.。

M2—Elevating motor of arm.

M3—Hydraulic pressure motor

M4—cooling pump

Caution：Must not turn the arm in one direction continuously to avoid twisting off wires due to there is not control ring on the top of the column.

9.2 Setting：

9.2.1 Current protector：20A breaker.

9.2.2 Check the phase sequence of machine

The headstock will loosen when pushing the headstock looseness button, which indicates the phase sequence of machine is right, otherwise adjust the position of random two power wires. Adjust the phase sequence of elevating motor after

adjusting the phase sequence of main power.

9.2.3 Demand space that operator repair the machine: Distance from the machine is 800mm.

9.3 Circuit explanation（refer to electric chart）

9.3.1 Preparation before starting the machine：

Open the door of electric box on cross arm，then switch on the breaker QF2、QF3，close the door finally.

9.3.2 Start：

Switch on the main switch under the column，power light HL1 will light when the current is switched on.

9.3.3 Run of spindle motor

Push the start pushbutton SB3，AC contactor KM1 will close and self-lock, spindle motor turn; push the stop pushbutton SB2, AC contactor KM1 release，spindle stop turning.

Set thermal relay FR1 to avoid overload running of spindle motor. The value is set by according to the current on the name plate of spindle motor.

9.3.4 Rise and fall of arm

Push up（or down）button SB4（or SB5），contactor KM4 will close，motor M3 clockwise turns，pressure oil enters arm loose oil cavity by distributor to push piston and rhombic block, which make the arm loosen, at the same time, the piston pushes the limit switch SQ2 by spring reed to make AC contactor KM4 break, and make AC contactor KM2 close, motor M3 stop turning, motor M2 circumvolve to drive the arm up or down.

If the arm does not loosen, limit switch SQ2 cannot close, AC contactor KM2 will not close, so arm cannot rise or fall.

When arm reaches target position, it will stop rising or falling when loosening the pushbutton SB4 (or SB5).

AC contactor KM5 closes 1.5 second later, motor M3 counterclockwise turn to supply the pressure oil. Pressure oil enters the clamp cavity to make the arm clamp, at the same time, piston pushes the limit switch SQ3 by spring reed to make switch off the AC contactor KM5, motor M3 stop turning.

Limit switch SQ1（SQ1a、SQ1b）is to limit the travel of arm, limit switch SQ1（SQ1a, SQ1b）acts when arm reaches target position, AC contactor KM2 breaks, motor M2 stop turning, arm stops vertical movement.

The automatic clamp of arm is controlled by the limit switch SQ3, and there are several conditions which the hydraulic pressure motor will overload run for a long time, such as there is malfunction on the hydraulic pressure system's clamp function, or limit switch SQ3 is adjusted improperly, which doesn't make the touch spot disconnect after clamping the arm.

Set thermal relay FR2 to avoid overload running of hydraulic pressure motor. The value is set by according to the current on the name plate of motor M3.

9.3.5 Looseness or clamp of column and headstock can be done separately or synchronously.

9.3.5.1 Synchronous looseness or Clamp of column and headstock.

Turn change-over switch SA to the middle position, at this time push the loose button SB6, electromagnet YV1, YV2 attracts, AC contactor KM4 attracts, motor M3 clockwise turn to make pressure oil enter loose oil cavity to drive piston and rhombic block, which make column and headstock loosen.

9.3.5.2 Looseness or Clamp of column and headstock can be done separately

Turn change-over switch SA left, at this time push the loose button SB6, electromagnet YV1, YV2 attracts separately, AC contactor KM4 attracts, motor M3 clockwise turn to make pressure oil enter loose oil cavity to drive piston and rhombic block, which make column and headstock loosen separately.

9.3.6 Start-up and Stop of Cooling pump

Power of cooling motor can be connected and disconnected by turning the switch QS2.

9.3.7 Emergency stop and release

Push Emergency stop button SB1, the machine stops, turn the Emergency stop button SB1 as the direction of arrow to reset.

Caution: Some of the electrical parts are still electriferous when pushing Emergency stop button SB1, the machine is non-electrified only after turning off the main power switch QS1.

9.3.8 Stop the machine：

Must switch off the main power switch QS1 before repairing the machine.

9.4 Maintenance and adjustment of electrical equipment

Caution：a、Be careful, the wire of the main switch and contactor still is electrified during checking the electrical equipment.

b、We suggest that the operator must turn off the switch QS1 before opening the electrical door.

9.4.1 Adjustment

9.4.1.1 Angle of elevating limit switch touch spot is adjusted well before leaving the factory, the elevating switch failure or arm doesn't act after pushing the button, open the cover of the limit switch if affirm the switch SQ1 has fault, adjust the angle of touch spot properly, then close the cover and screw down the bolts.

9.4.1.2 Adjustment of micro switches SQ2、SQ3

Position of micro switches SQ2, SQ3 are adjusted finely. When fault happens in vertical direction, remove end cover on the left of cross arm if find cross arm is not loose, or clamp cannot reset; adjust the position of micro switches SQ2, SQ3 if the position is not reasonable, fix the adjustable plate well finally.

9.4.1.3 Setting the current value of thermal relay

The value is set by according to the current on the name plate of motor.（please refer to details of electric）。

9.4.1.4 resume running state of PLC：Under normal conditions vertical line moves back and forth on the screen after connecting the power，which indicates PLC is running, otherwise do as the following steps：push ESC→ appear words on the screen, then push the arrow→run→OK 2 times→ESC（return）

How to judge PLC program loses：after resuming running state of PLC, push the loose (clamp) or up (down) button, but the machine does not act, which indicates the data of PLC has been switched off, it need to exchange the PLC with new PLC with program.

9.4.2 Preventive check

To be more safety，the electrical equipment should be inspected yearly, eliminate

it if checker finds fault at once.

9.4.2.1 Measurement of insulated resistance

Measure main loop and control loop with 500V Megohmmeter，the insulated value is greater than 1 megohm.

9.4.2.2 Check of ground protection

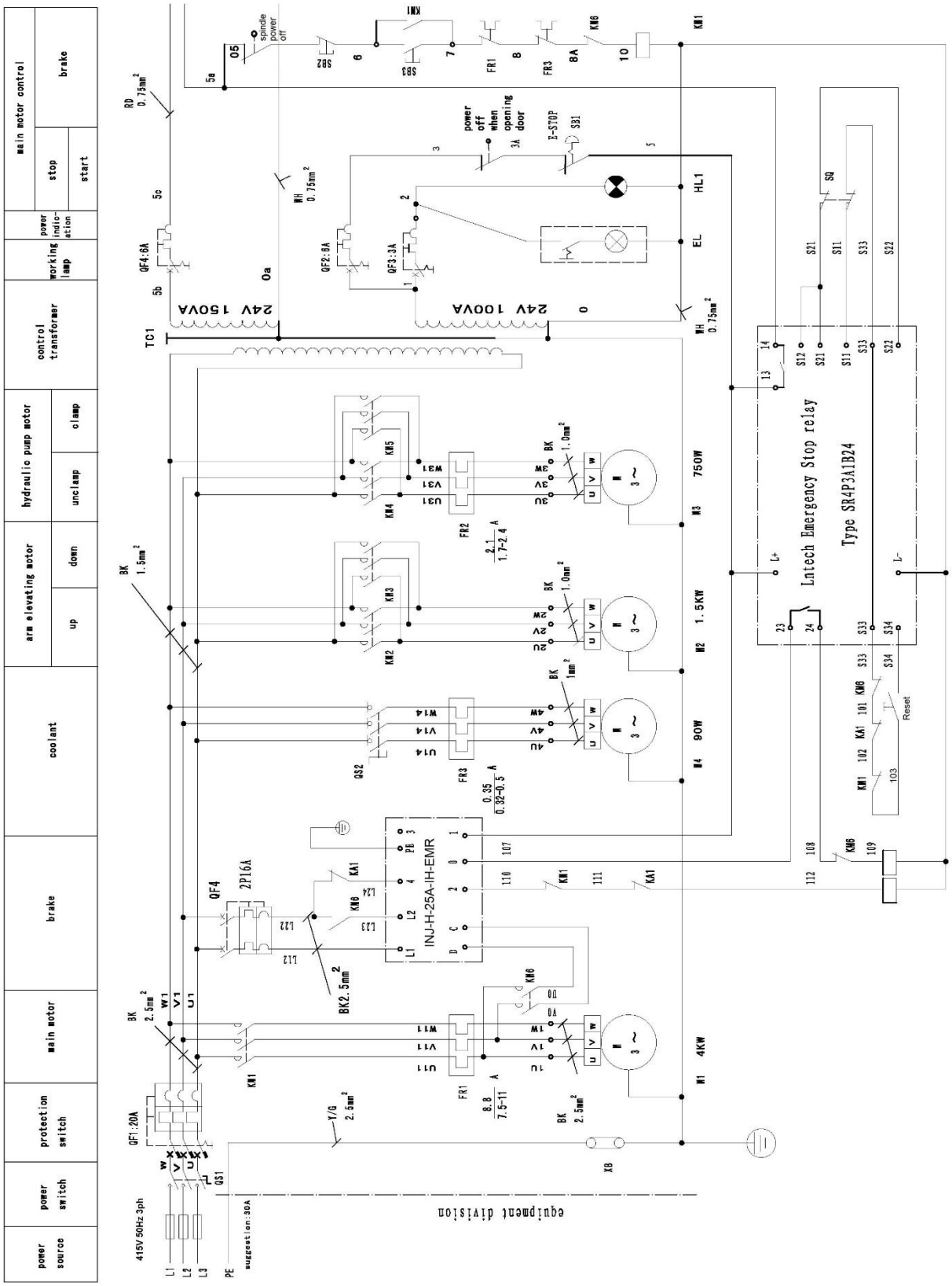
There is ground protection on all motors, operation panel and the electrical door, so please check whether the ground screws are screwed down.

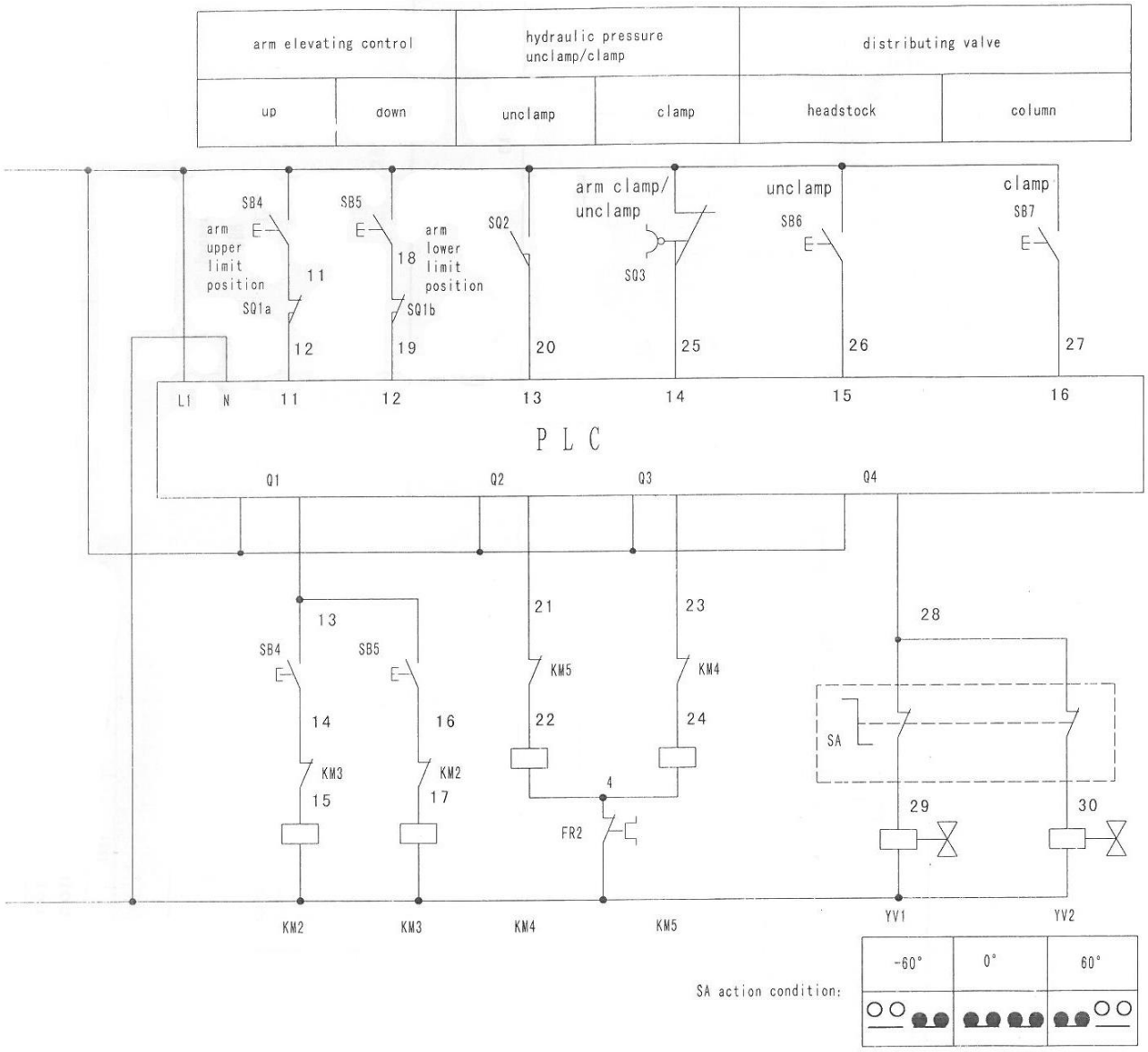
9.4.3 Troubleshooting of electric：

Search for and eliminate fault according to wiring diagram and electrical components.

Symptom	Possible cause	Remedy
Spindle does not rotate.	1. The switch disconnects. 2. Thermal relay FR1 overheat or trip. 3. Low voltage. 4. Control loop has fault 5. Motor has fault. 6. Short circuit in line cord or plug.	1. Connect the switch. 2. Reset relay by pushing the reset button after it is cool or replace the new one. 3. Check power supply for proper voltage. 4. Check control loop and eliminate fault. 5. Check motor and eliminate fault or replace the new motor. 6. Inspect cord or plug for insulation and shorted wire and replace extension cord.
Elevation of cross arm has fault.	1. The switch disconnects. 2. Control loop has fault. 3. Motor has fault.	1. Connect the switch. 2. Check control loop and eliminate fault according to wiring diagram. 3. Check motor and eliminate fault or replace the new motor.

Clamp or looseness of hydraulic pressure has fault.	1. The switch disconnects. 2. Thermal relay FR2 overheat . 3. Control loop has fault. 4. Mechanism or oil way has fault.	1. Connect the switch. 2. Reset relay by pushing the reset button after it is cool or replace the new one. 3. Check control loop and eliminate fault according to wiring diagram. 4. Check mechanism or oil way parts and eliminate fault.
Cooling system has fault	1. The switch disconnects. 2. Control loop has fault.	1. Connect the switch. 2. Check control loop and eliminate fault according to wiring diagram.


power source
power switch
protection switch
main motor
brake
coolant
arm elevating motor
up
down
hydraulic pump motor
unclamp
clamp
control transformer
working lamp
power indication
main motor control
start
stop
brake
415V 50Hz 3ph
L1
L2
L3
PE
suggestion:30A
equipment division
QF1:20A
QS1
QF4:6A
QF2:8A
QF3:3A
TC1
24V 150VA
24V 100VA
spindle power off
power off when opening door
E-STOP
S31
HL1
EL
INJ-H-25A-IH-EMR
QF4
2P16A
KM1
KM2
KM3
KM4
KM5
KM6
KA1
FR1
FR2
FR3
QS2
M1 4KW
M2 1.5KW
M3 750W
M4 90W
Intech Emergency Stop relay
Type SR4P3A1B24
Reset
XB


arm elevating control
hydraulic pressure unclamp/clamp
distributing valve
up
down
unclamp
clamp
headstock
column
SB4
arm upper limit position
SQ1a
SB5
arm lower limit position
SQ1b
SQ2
arm clamp/ unclamp
SQ3
unclamp
SB6
clamp
SB7
11
12
18
19
20
25
26
27
L1
N
11
12
13
14
15
16
PLC
Q1
Q2
Q3
Q4
13
SB4
14
KM3
15
SB5
16
KM2
17
21
KM5
22
23
KM4
24
4
FR2
28
SA
29
30
KM2
KM3
KM4
KM5
YV1
YV2
SA action condition:
-60°
0°
60°

Main Electrical Components List

Code	Name	Specification	Qty	Remark
M1	Motor	Y112M-4 415v 50Hz 3PH 4kW V1	1	
M2	Motor	Y90L-4 415v 50Hz 3PH 1.5kW V1	1	
M3	Motor	YSJ80-4 415v 50Hz 3PH 0.75kW V1	1	
M4	Cooling pump	AOB-25 415v 50Hz 3PH 90W	1	
TC	Control transformer	JBK5-160 160VA I:0-415V O:0-110V(110VA)、 0-24V(50VA)	1	
QS1	Power switch	JFD11-25/31 red	1	
QS2	Cooling switch	JFD11-25/31 black	1	
QF1	breaker	DZ47-63(3P 20A)	1	
QF2 QF3	breaker	DZ47-63(1P 3A)	2	
FR1	thermal relay	JR29-25(7-10A)	1	
FR2	thermal relay	JR29-25(1.6-2.5A)	1	
KM1	AC contactor	CJX2-1810(AC 110V)	1	
KM2-KM5	AC contactor	CJX2-0901(AC 110V)	4	
PLC	Intellectual controller		1	
SQ1	Limit switch	HZ4-22	1	
SQ2、SQ3	Limit switch	LX5-11	2	
EL	Working light	JC25(AC 24V 40W)	1	
HL	power light	AD11-22/20(AC 24V) white	1	
SB1	E-stop pushbutton	LAY7-01ZSM/1 red	1	

Code	Name	Specification	Qty	Remark
SB2	Pushbutton	LAY7-01BN/1 red	1	
SB3	Pushbutton	LAY7-10BN/2 green	1	
SB4-SB7	Pushbutton	LAY7-11BN/3 black	4	
SA	Knob	LAY7-02X/3102	1	
YV1-YV2	electromagnetic valve	MFJ1-3(AC 110V)	2	